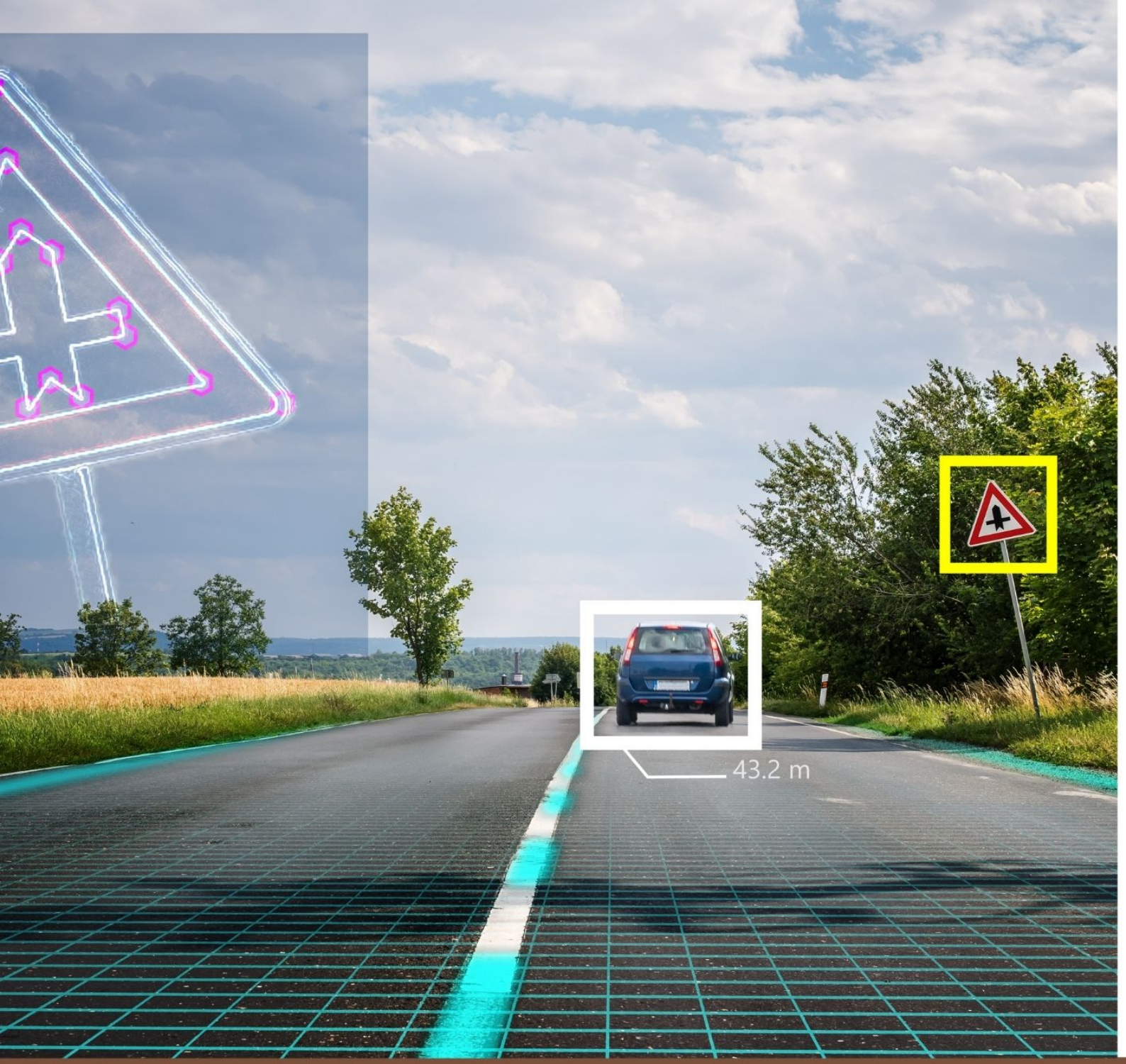


COMPUTER VISION

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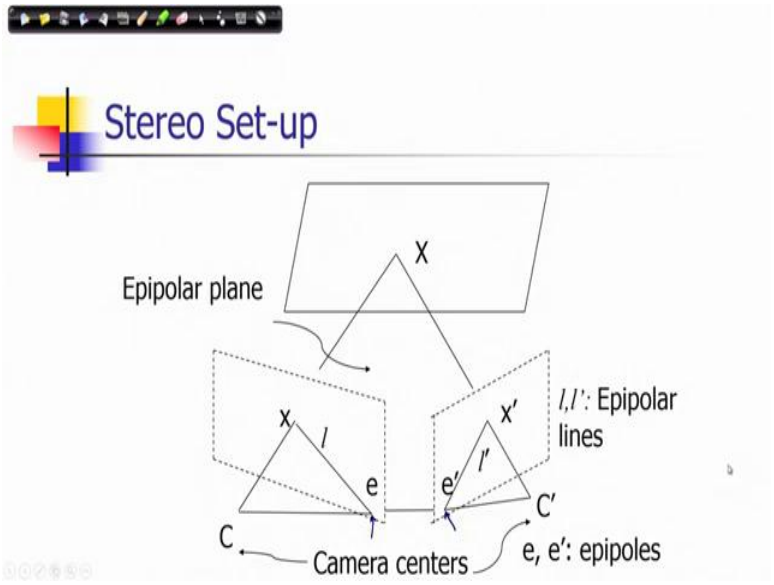
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Lecture - 16
Stereo Geometry Part – 1

In this lecture, we will start a new topic that is on Stereo Geometry. So, let us consider what is meant by stereo setup. In the stereo setup, we have two cameras and in this two cameras, they are specified by their camera centers and their image plane.

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And scene point X for which you will be taking the images in both this cameras. So, if I apply the rule of projection here. So, we get the image of this scene point X in the first camera, whose camera center is given as C as a say x.

Similarly the second camera given by the camera center C' forms an image of that same scene point at in its own image plane. So, you have two images of the same scene point and this is what y x' ou will get in a stereo setup. For every scene point, there are two image points, now if they are viewable from the two cameras itself.

Now, this particular construct has certain properties geometric properties. What we can see particularly that the point C, X, C' and the if I consider a plane found by C C' and the scene point X. So, the image points x and x', they also lie on the same plane.

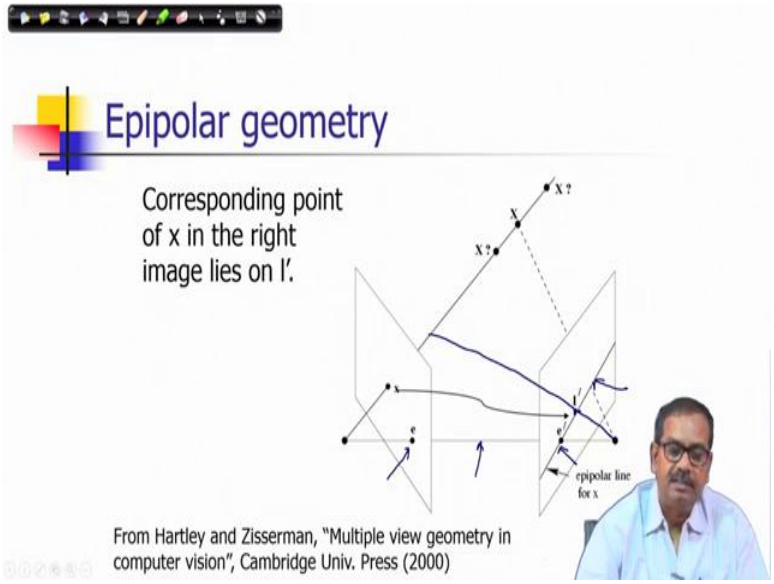
There is a from first restriction, also if I draw a line between C and C' which is called for a stereo setup as the base line. So, that also lies on the same plane; that line intersects the image planes..

Again, there are some image points in the image plane and in fact, as we can see that one of these intersection point is here which can be interpreted as the image of camera center of the other camera. And the other point is shown here point of intersection of the base line with the other image plane, which we can interpret also as image of camera center of the first camera.

So, these two points, these are called as epipoles and they are also lying on the same points. So, they are denoted here by e and e' and the line joining e and x which is nothing but intersection of the plane containing all these points X, C, C' and their image points. So, that intersection with the first image plane, it will be giving you another line and similarly get two such lines in two image planes because the intersections with respect to the three-dimensional plane formed by same point and camera centers.

Now, those two point; those two lines they are called epipolar lines. So, we have this particular configuration and these terms are used we will be explaining these terms more and more in subsequent lectures. But you should note this interesting configuration that all of this points and lines, they lie on the same plane and that plane is called epipolar plane with respect to this same point. So, you have this epipolar lines and this plane is called epipolar plane, which is formed by the camera centers both the camera centers and a same point.

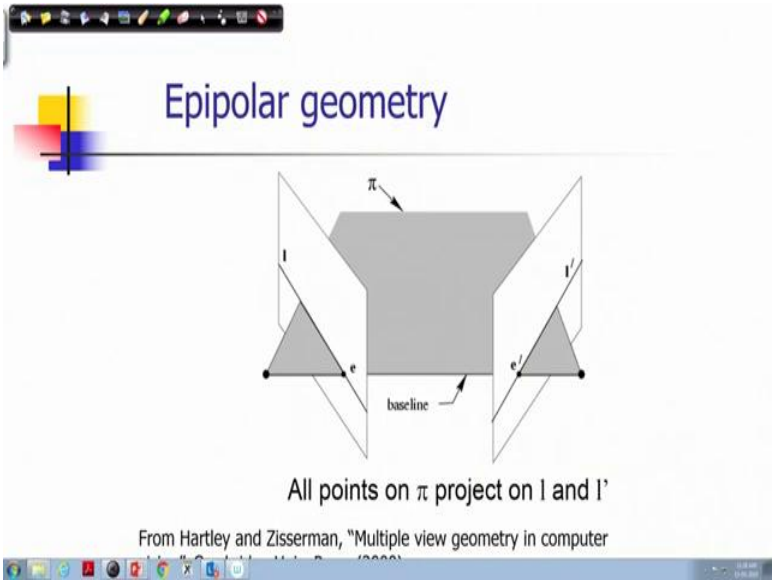
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So, this geometry in general is referred as epipolar geometry. So, that summary of this again; once again different concepts of the epipolar geometry is shown here. As you can see that you have the corresponding base line here and the intersection of base line is giving in the epipoles and you note particularly this epipolar line in the second camera. So, if I consider the projection ray formed by the first camera and the image point, then any point lying on that ray is a possible candidate of a same point from which whose image is this x and if I take the image of the scene point by the second camera.

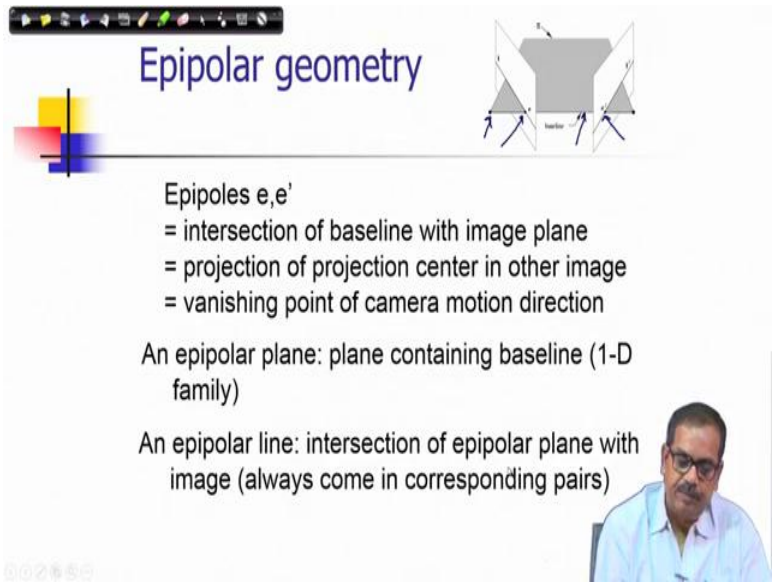
So, this image would lie on the epipolar line. So, that is a geometric interpretation. who all the possible candidates of same point with respect of this image point, they lie on an epipolar line. So, this is a constraint that is imposed by this particular configuration. So, corresponding point of X in the right image lies on l prime.

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And you can also on interpret this configuration in this fashion also for any epipolar plane π, its point of intersections are those two; its line of intersections with respect to image plane. Those are the corresponding epipolar line. So, all points on π the project on those lines only. So, any point on plane π will be projected for image for the first camera will be projected on l; epipolar line l and for the second camera, it will be projected on epipolar line l'.

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So, in an epipolar geometry we have this concepts once again we will be now defining them in particular using the geometry configurations only. So, epipoles as we see e and e' in the figure here. So, epipoles are defined as the intersection of baseline with image plane that is one kind of definition. So, as you can see that this is a baseline, this is a baseline and intersection with image planes respectively, they provide you those points of they are the epipoles. That is one kind of interpretation or you can also interpret as I mentioned earlier, that projection of projection center in other image because you can consider for example, this is the camera center which is a projection center of the first camera.

So, these epipole is the projection of this camera on this image plane because you draw the ray from this point to this point which is again the baseline and its point of intersection of the image plane will give you these particular epipole. We call this epipole as a right epipole. So, in our convention we will have this particular directionality, we will be using this directional notations. So, first camera we will call left camera from the diagram's representation . So, you will also sometimes refer it as left camera and the second camera, we will refer it as right camera.

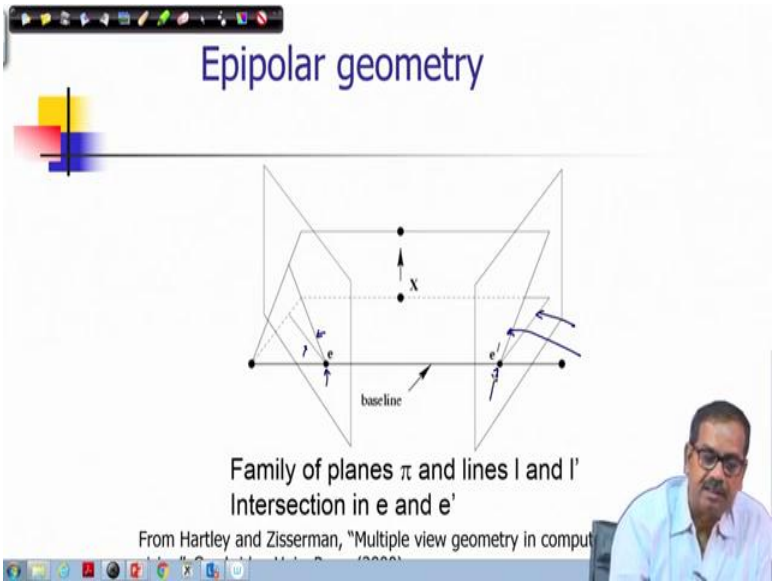
Similarly, the epipoles out of this first epipole e is considered the left epipole, that is just the epipole on the reference image plane or first image plan. So, first camera is considered as a reference camera of the stereo setup and second epipole, it sometimes it is also referred as right epipole that is on the corresponding epipole on the second camera. So, these are certain notations also or terminologies will be used in our discussion.

You can also interpret epipoles as a vanishing point of camera motion direction. So, what is a camera motion direction? It is a direction of translation of the camera center. So, which is again given by the baseline. So, in that direction as you can see that it is the intersection of that ray with respecting image plane will give you that intercepting point will be always the vanishing point along that direction. So, epipoles are also vanishing point of camera motion direction. So, in various ways you can interpret this epipoles.

On the other hand, you define an epipolar plane it is the plane containing base line. So, you can see that baseline. So, it is an one dimensional family in the sense that it is a pencil its baseline is acting like a pencil and you know all the planes over centering

around that particular line is defining an epipole; epipolar plane. And epipolar line is a intersection of epipolar plane with image and it always come in corresponding pairs.

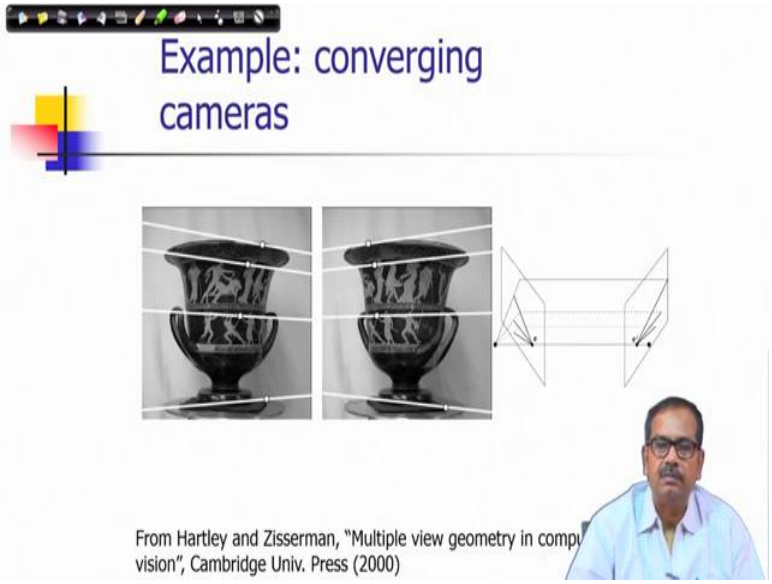
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So, this is a explanation that was mentioning that you get a family of epipolar planes, intersection of each any pair of the plane is a baseline. And this epipolar planes that intersects with the image plane to provide you epipolar lines and all those epipolar lines, they are also meeting at epipoles in respective epipoles that we can see in this particular diagram.

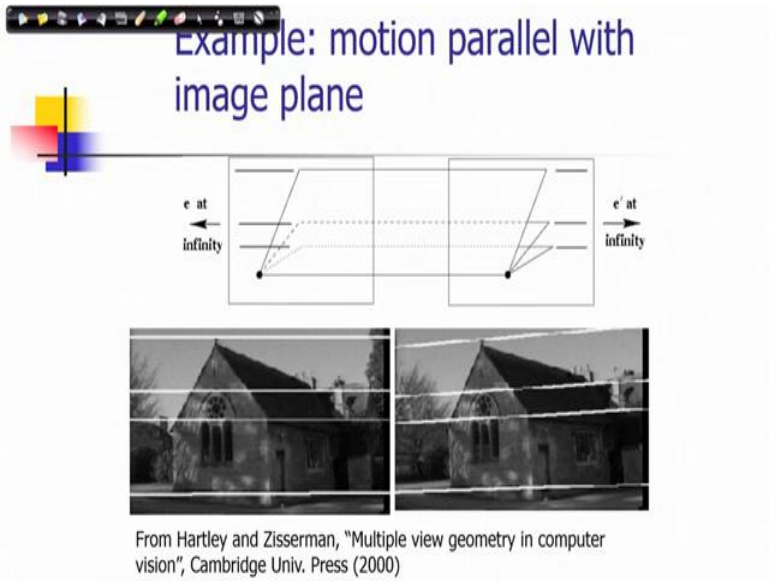
So, you can consider. So, this is a plane and this is a another plane. And for each plane, now these are the corresponding epipolar lines and they are intersecting at e in this case were intersecting at e'.

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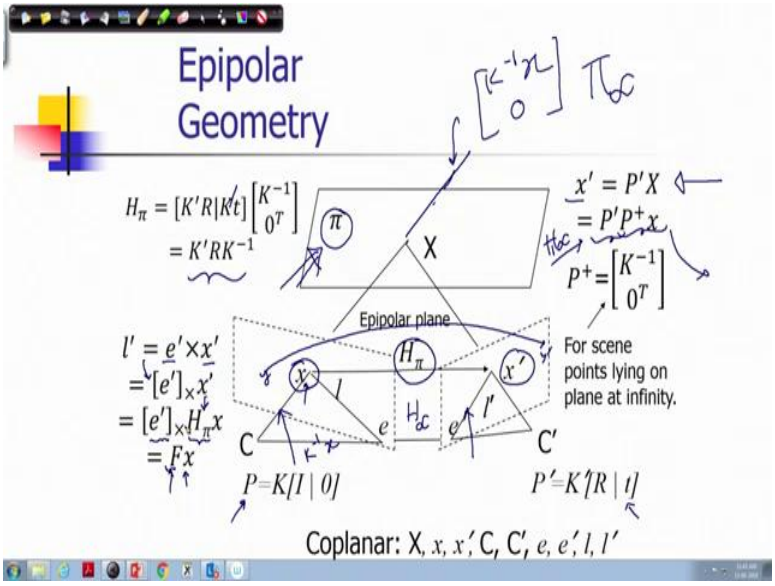
So, typical examples of converging camera images. So, were it is shown that their epipolar lines are converging. So, this is one example, I have taken from the book of Hartley and Zisserman which is also shown here multiple view geometry in computer vision. You can find, you can see that how epipolar lines they can be mapped on those images and you can find out those all those epipolar lines, they are converging in this case.

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Also you take a situation, when you have simply translation of camera center. There is no rotation of axis. So, which means your epipolar lines they become parallel so, all epipolar lines they become parallel; parallel to the direction of motion of the image planes or the camera. So, since epipolar parallel lines, they meet at infinity. So, this epipoles also in the image plane, they are the points which are lying at line at infinity. So, these are the interpretations of epipoles when you have simply a translation of camera centers.

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So, now let me provide you the mathematical formulation of different concepts involving these different entities which we have defined like, first thing as we can see from this geometry. Once again, that you have two cameras given by the centers C and C' and also the projection matrices of the each camera, those are also specified here. So, one of the projection matrix that is given as $K[I \mid 0]$; so that means, it is a reference camera and it is a camera centric coordinate system that we considered here. And on the other hand, the other camera that is the second camera is given by this ($K'[R \mid t]$) K' and then, the rotation and translation form as R and t.

So, the in a inwards studio set up, the reference camera is denoted in this diagram in this form. So, one of them is a reference camera and the other one is a second camera. This is a notation or left camera usually we put the reference camera call it as a left camera and the right camera as the other camera with respect to the reference camera. Its coordinates

are expressed and reference camera is usually they have this camera centric coordinate system, but it could be also generalized.

Now, you can see that in this structure, we have two image points corresponding to the same points that is x and x'. So, we can actually show that given all the points in this plane, it induces a homography. We will discuss it later on, but right now we will consider a particular type of homography, we will consider the mathematical relationships and show that how that homography exists between corresponding points.

So, you know this particular relations up camera geometry that x' can be expressed as $P'X$ in the homogeneous coordinate system in the projective geometry. And then, we can consider the ray found by this particular image point with respect to camera center and consider a point which is lying at infinity along that ray. So, that is expressed as pseudo inverse P plus x ($P^{+}x$).

So, I should not call exactly that is pseudo inverse, but it is given in this structure $\begin{bmatrix} K^{-1} \\ 0^{T} \end{bmatrix}$.

You can note that the direction cosine, direction of the particular line, this can be given as $K^{-1}x$ and the any point which is lying at infinity for this line that can be expressed as $\begin{bmatrix} K^{-1}x \\ 0 \end{bmatrix}$ ok. So, finally, this is a same point which we were considering which is at a infinity and we are considering an image of that point in this particular configuration.

So, if I consider for every image point for any other image point say y, we have also there is a corresponding image point y' which is again formed by the scene point which is lying at that infinity and that plane is called plane at infinity. So, there is a particular notation for that and all these points. So, this plane let us assume in this case that we will only considering the formation of corresponding points with respect to plane at infinity, then we can get an interesting homography between those corresponding points.

That means, , the point in the first camera first image plane and the corresponding points which are found from plane at infinity from the same points of the plane at infinity the second camera and this particular homography, we also sometimes referred as homography at infinity. So, this discussion also we will be doing later on, but let this structure of at infinity simple because as you can see from this relation x' and this is

nothing but it is giving you the corresponding H_α and which is easily computed in this form ($P'P^+$).

So, this is what scene points lying at plane at infinity. So, I can simplify this matrix by its corresponding elements. So, this P' is given by $K'R$ and it should be $K't$. So, if I multiply, then you get the homography matrix as $K'RK^{-1}$. So, this is your point x', now the these line is reconstructed as I know in the projective space, it should be cross product of e and x' that would give you this line $l' = e' \times x'$.

Now this operation these are all three vectors and we know how to compute this cross product of this operation also can be expressed in terms of matrix multiplication. I form a three dimensional matrix with the elements of e' and I can express this particular cross product as I mentioned that with respect to multiplication of that matrix. So, let me provide you a simple notation.

So, just to complete this discussion, we will come back to these particular constraints. Suppose there is a matrix let us say 3×3 matrix and it is multiplied with x prime that is giving you l prime which is performing equivalent computation of this cross product of 2 points. So, I can then expand this operation. So, this x' can be expressed as also the using the homography relation in this form; $H_\pi x$ and since this is a 3×3 matrix, this a 3×3 matrix, again I can multiply this two matrices in composite form and let me denote it by F ($[e']_\times H_\pi$).

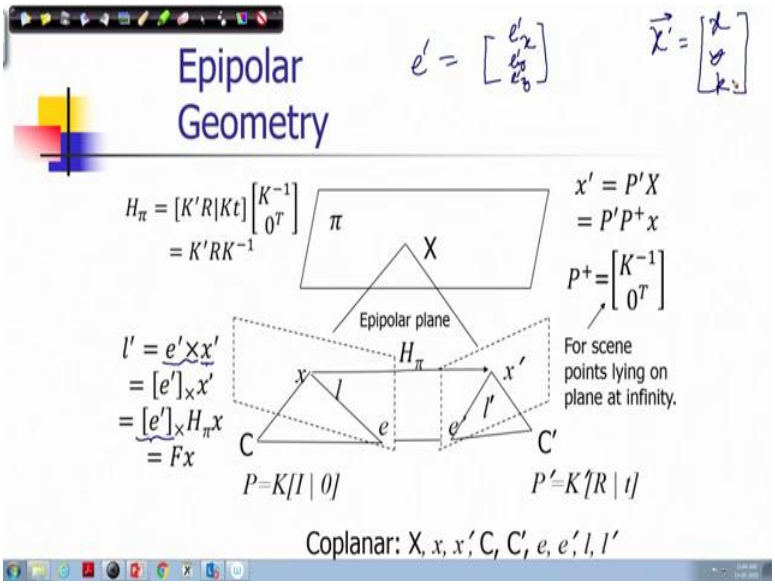
So, this relation l' this epipolar line in the second camera can be obtain from the image point in the first camera. It is a very important relationship in this stereo setup; is a very fundamental relationship in a stereo setup and that is they are related by this particular

Expression ($l' = Fx$). So, there exist a matrix which can be obtained from this you know from this parameters, from this values and if I multiply the image point in the first camera in its homogenous coordinate system, with this matrix F which is of dimension 3×3 , then I will get this epipolar line l'.

So, this matrix is called fundamental matrix and that is a very characterizing property or characterizing matrix of this stereo setup and its task is that it converts an image point to a line which is an epipolar line and sometimes they are called also, this relationship is

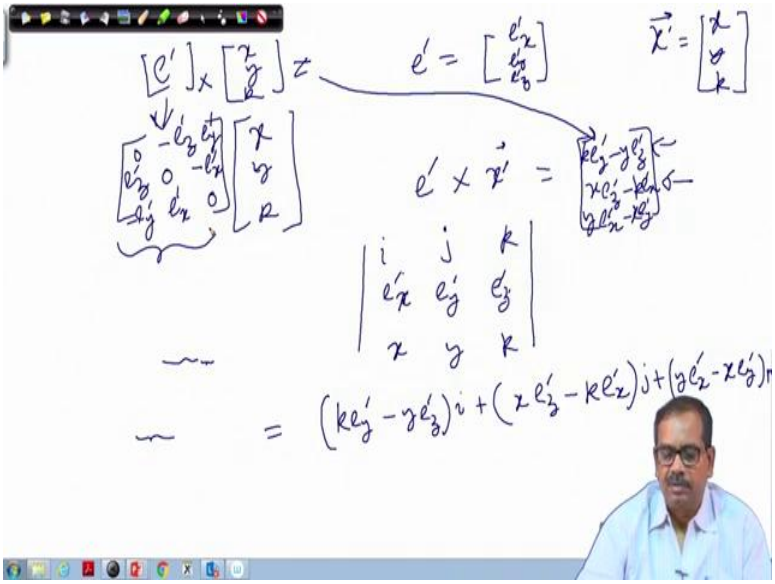
also called co-relation. This transformation is called co-relation. That is a kind of misnomer, but you may find in the text book this term is also there.

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So, let me also explain how this particular conversion means happening here, that a cross product can be expressed as the form of a three-dimensional matrix multiplication. So, if I consider e'. So, let us consider e' is represented by the column vector say $\begin{bmatrix} e_x \\ e_y \\ e_z \end{bmatrix}$ say let us consider these are the column vectors. And a point x' is: as you know that this is a basically a column vector ($\begin{bmatrix} x \\ y \\ k \end{bmatrix}$) and is its coordinate, let me represent it as say x y and some scale value k that is the homogenous coordinate system. So, let me represent in this form.

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So, when we have a matrix multiplication; so, let us consider this scenario. So, when you have this , not the cross product of $e' \times x'$. So what we will do?

$$\begin{bmatrix} i & j & k \\ e'_x & e'_y & e'_z \\ x & y & k \end{bmatrix}$$

$$(ke'_y - ye'_z)i + (xe'_z - ke'_x)j + (ye'_x - xe'_y)k$$

We will consider this i j k and this is e x prime e y prime e z prime and this is say x y k. So, I need to compute this expression. So, let me expand this expression as we did earlier also. So, we can write it as say k e y prime minus y e z prime, this is i plus then x e z prime minus k e x prime, this is j plus y e x prime minus x e y prime, this is k. So, you get this particular expression. So, you would like to have a vector which give you all these things; that means, it should give you this is

$$\begin{bmatrix} ke'_y - ye'_z \\ xe'_z - ke'_x \\ ye'_x - xe'_y \end{bmatrix}$$

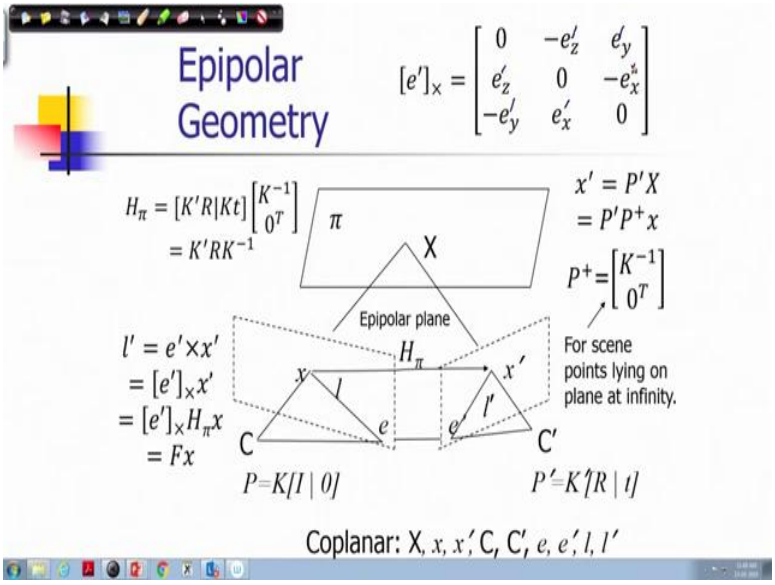
k e y prime minus y e z prime, then x e z prime minus k e x prime and y e x prime minus x e y prime.

So, I should have a matrix with respect to in a matrix multiplication, as I mentioned that we are considering replacing the as a matrix multiplications. So, it should multiply with $\begin{bmatrix} x \\ y \\ k \end{bmatrix}$ which should give me this particular expression. So, I have to find out this value.

Now, you can see that to get this what I have to do? So, there is no x component in this row. So, this should be 0, then for y it is $-e'_z$; for k it is e'_y. Consider the second row so, the x contains e'_z, then this is 0 and then with k, it is $-e'_x$ and consider the third row, where x contain $-e'_y$, then y e'_x and this is 0. $\begin{bmatrix} 0 & -e'_z & e'_y \\ e'_z & 0 & -e'_x \\ -e'_y & e'_x & 0 \end{bmatrix} \begin{bmatrix} x \\ y \\ k \end{bmatrix}$

So, this is what is a you know conversion of $e' \times x'$, I can as well replace it by the three-dimensional multiplication. You note particularly these matrix is a skew symmetric matrix. Its diagonal all the diagonal elements are 0 and you can find out the respective transpositions will give you a negation of the particular matrix elements. So, this is how this is explained in this case.

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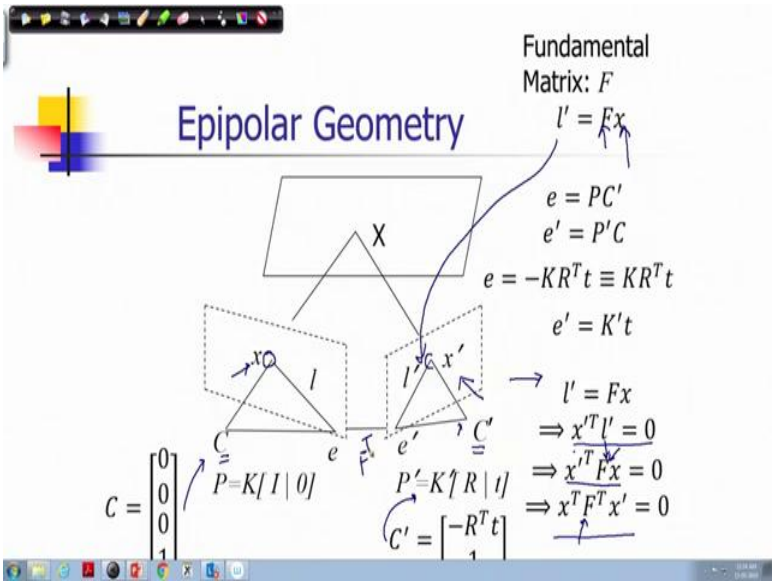


So, we will proceed with this particular explanation that how it could be converted and as I mentioned earlier that how this relationships, matrix relationships can be expressed into form of a skew symmetric matrix. Now you can use prime(‘) in my notation, I have

used prime(‘) every case because they are all related to e'. However, you know it is just a matter of notation that you need to be used to.

So, fundamental matrix as you can see the expanded form, when I am considering only the elements of projection matrices and the epipoles that can be written as $[e']_{\times} K' R K^{-1}$. So, I can express this fundamental matrix by the camera matrices and epipolar configurations, from the epipoles itself.

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So, we will continue this discussion on epipolar geometry, once again that fundamental matrix is the matrix, if I multiply any image point in the projective space in the first camera, if I multiply that with this matrix F. Then, I will get the epipolar line in the second camera. So, the interpretation is that the image point so, all my corresponding image point any or rather corresponding image point will lie on this epipolar line. , which means that I can apply the point contentment relationship with line, you note that this center of the camera is given by this, center of the first camera is the given by this one and center of the second camera is also given by this particular expression. So, this is $\begin{bmatrix} -R^T t \\ 1 \end{bmatrix}$ so, given by this expression.

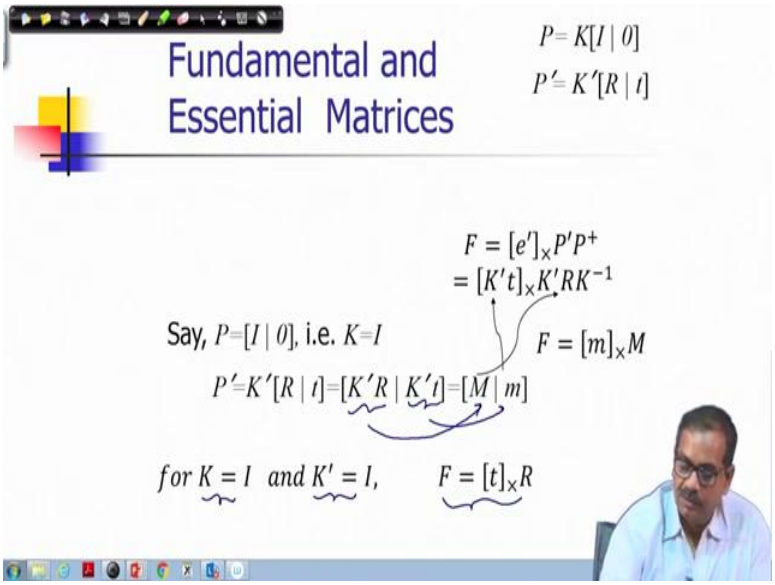
And as we have discussed that epipole can we considered as image of the camera centers. So, for the left epipole e, it can be expressed as if I multiply the projection matrix with

the C', I will get the image coordinate of e and which is given by $-KR^T t$. So, which is $KR^T t$ or this will equivalent because you know directionality does not matter in this case; minus sign just denotes the directionality. Similarly, e' can be considered as the image of the image of the first camera center e' which is $K't$. So, e is image of C', e is image of C' and e' is image of C and this is how the relationship is there and the point contentment relationship with the epipolar line that is expressed here.

So, I can get the epipolar line l' that can be expressed as Fx and if I apply the point contentment relationship $x'^T l' = 0$ and expand l' in the form of equal to Fx. So, we get a relationship $x'^T Fx = 0$. So, given two corresponding points which are observable or measureable or computable with two images; that means, they are the points of the same point. Then, their existed matrix 3×3 matrix which satisfies this relation.

So, this relationship is also a very fundamental to epipolar geometry and infact. Now if I apply the matrix transposition rule, I can also express this is as $x' F^T x^T = 0$ that means, if I consider this is a reference camera and this is a second camera. So, with respect to that point correspondence also we have a matrix which satisfy that relationship and those matrices are called fundamental matrix and they are related by this relation.

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So, if this is a reference camera with respect to this camera, if the fundamental matrix is F, if I consider the other camera is reference camera, then the fundamental matrix will be

F^T . That is what we get in this relationship. There is another particular term which is used, terminology which is used in epipolar geometry when the camera matrix when the camera is a calibrated camera.

So, we say a camera is calibrated, when the calibration matrix is known to us and then, we see that how this relationship can be simplified. So, when you have a calibration; when you know the calibration matrix, I can express any image coordinate into its normalized canonical coordinate system.

So, you can see that the relationship between the fundamental matrix and the camera parameters is given in this form ($F = [e']_\times P' P^+ = [K't]_\times K'RK^{-1}$). Given the two cameras as P and P' as shown here and suppose P is given as simply in a canonical form ([I|0]), normalized canonical form the calibration matrix itself ; that means, we have a apply all necessary transformation to bring it into the standard pinhole camera geometry; the base pinhole camera geometry configurations with the base coordinate convention of cameras centric coordinate system.

So, if I apply once again the representation of P' in this form, then this relation can be expressed as so, you can see that K'^T can be also considered as this is a representation.

$$P' = K'[R \mid t] = [K'R \mid K't] = [M \mid m]$$

This is an another representation of camera system and if I put the calibration matrices, I since if I know the calibration matrices I can always covert them by doing the necessary transformation.

Then, this particular structure will relate to t. There is a translation parameters $[t]_\times R$. So, this is how the fundamental matrices expressed in that form and then, this fundamental matrix is called essential matrix. In this particular form, this fundamental matrix is called is essential matrix.

So, with this, I will stop my lecture, this lecture here and we will continue this discussion.

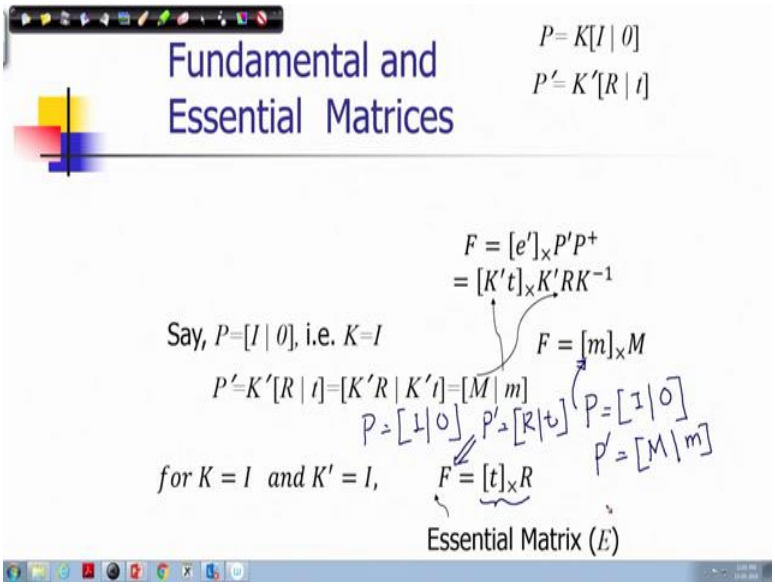
So, thank you for listening to this lecture.

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Lecture – 17
Stereo Geometry Part – II

We will continue our discussion on epipolar at geometry and we are discussing the definition we discussed about fundamental matrix and how this matrix is related with the parameters of projection matrices. First thing is that a fundamental matrix is a matrix if we multiply any image point it will give you the corresponding epipolar line in the second image plane.

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And this can be; this can be computed from the camera parameters itself and those relations are shown here which we discussed earlier also, it could be in various forms. Here you can consider that you have the reference camera is in the form of say standard canonical representation say [I|0]. And the other camera this is given in this representation, with respect to this we can compute the fundamental matrix from this configuration you can compute it by this expression.

So, this is one kind of relationship that we discussed and we also observed that a fundamental matrix can be simplified into a form of an essential matrix if I use the calibrated camera which means I know the calibration matrices. And then I can make

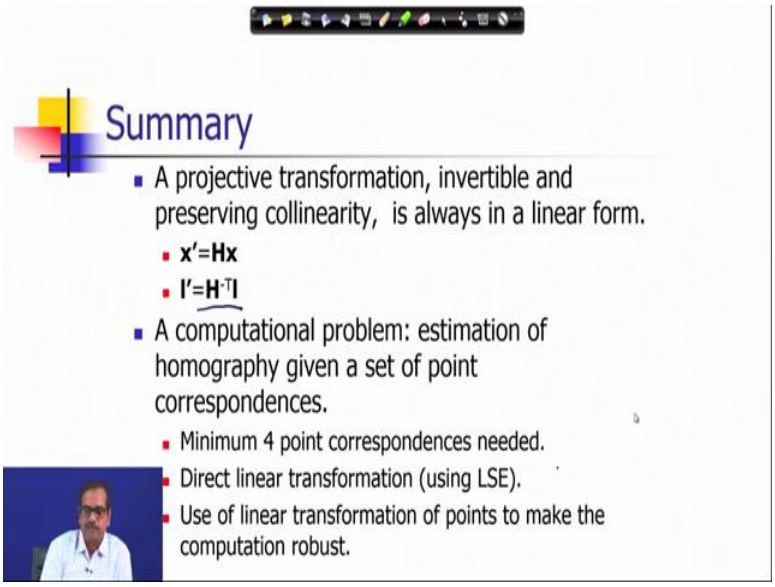
necessary transformations on the image coordinates so, that effectively calibration matrix becomes an identity matrix. So, in that form its fundamental matrix its reduces to this particular structure when you are defining the camera mattresses in this form. So, P is given as [I|0] and P' is given as rotation matrix and the corresponding, translation matrix translation of the origin.

So, they are given in this form. So, then the fundamental matrix can be obtained by performing this operation. So, it is a cross product of translation vector with respect to the rotation matrix. So, this is a relationship and we have also defined how this notation, how you can convert a cross product operations into this matrix multiplication from.

$$F = [t]_X R$$

And then this fundamental matrix is called essential matrix and we denote in this particular structure by another notation E in our discussion. So, this is what we discussed in the last lecture, now I will consider solving a problem just to illustrate how these concepts could be used to retrieve various information.

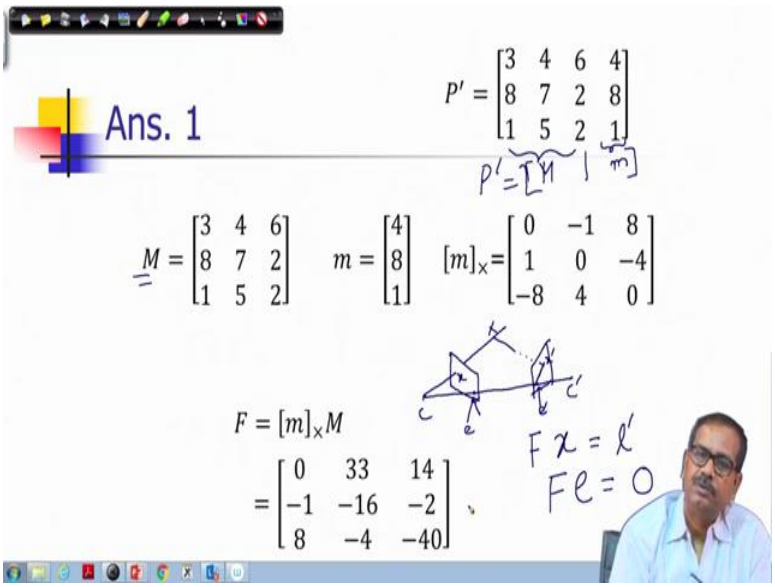
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So, take this problem, here in this case it is a problem of simply computing the fundamental matrix. So, you look at the problem once again that reference matrix P is given as [I|0] which is in the canonical form itself whereas, P' it is in the non-canonical form representation; that means, it is not calibrated. Its calibration parameters are not known to

us rather I know all the elements of the projection matrix that is what is P'. So, with this configuration we would like to compute the fundamental matrix of the system. So, let us see how this configuration we can do.

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So, this is given that P' as I mentioned P is also given here. So, here we will take this 3 X 3 sub matrix which is denoted here M and then we will also consider the column vector which is denoted the m. So, these are the notations we have used earlier. So, we represent this P' into this form into this notation with the sub matrices. Then we will apply the corresponding know relations between M and m which will are configuration involving M and m which will give us fundamental matrix.

So, we first will represent to the scheme symmetric matrix we will get the scheme symmetric matrix. So, which is performing the equivalent cross product operation with the vectors. So, from [4 8 1] we can get this scheme symmetric matrix and then fundamental matrix is given by these computation. So, if I multiply this matrix with M so, this m; m cross let me call this matrix as simply $[m]_X$ matrix. So, this $[m]_X$ matrix if I multiply with M, then we will get a 3 X 3 matrix.

So, this is what the fundamental matrix is. So, as you understand if the camera matrices in these particular forms then it is very easy to compute the fundamental matrix. We will also discuss that how a general forms of camera matrices can be also used for deriving this fundamental matrix. One thing you should note one of the property of the fundamental

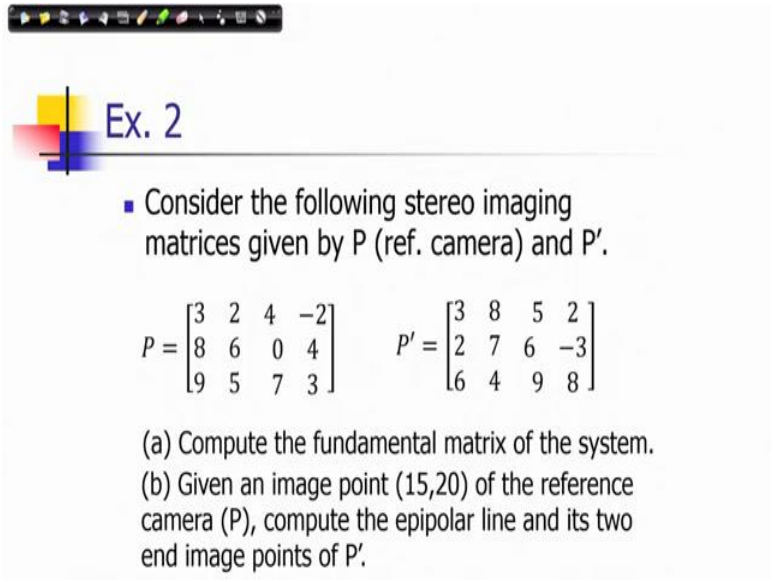
matrix that is there which I need to discuss here, that in a fundamental matrix if I perform as I mentioned if I convert consider any image point x. And if I multiplying with F then you get the corresponding epipolar line in the right plane, in the right image plane.

And what is an epipolar line? Epipolar line is a line found by the epipole in this configuration. So, you have this configuration. So, this is center C, this is center C' and this is the baseline. So, these are the epipoles, this considered these are the intersecting points. So, this is e' and a point x and its corresponding image point say this is x this is x '. So, epipolar line is formed by these two points; one is the epipoles in the image plane right image plane with or we call it right epipole or e' and the image point. Now, you just consider this image point incidentally which is the left epipolar e.

Now, its corresponding image point itself is e', then what would be epipolar line, how do you form this epipolar line? So, as we have seen earlier also that actually this is the case of singularity. You cannot form a line by just by connecting the same at the same point itself, you cannot define it on a line just using a single point. So, that is why this mathematically this is geometric constraint geometric constraint that we understand. But, mathematically this constraint is expressed in this form if I multiply this fundamental matrix with this epipole in the image point what I will get?

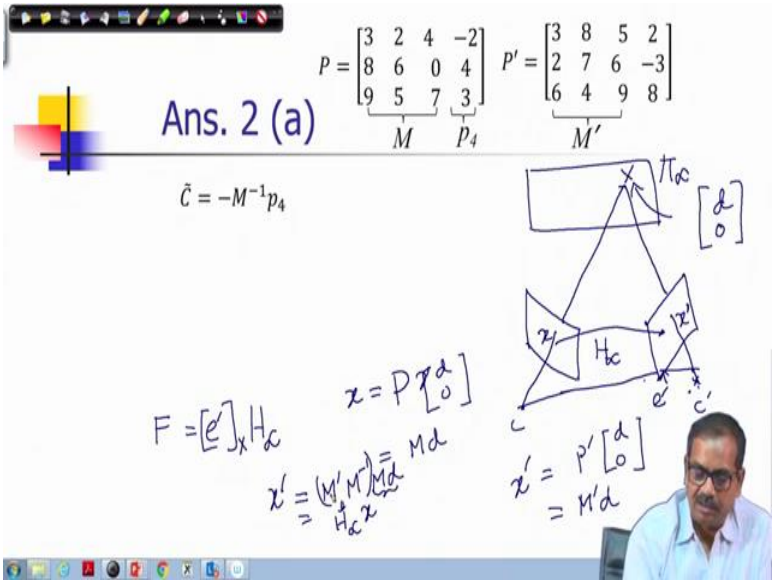
I will get a 0; that means, a 0 column vector. So, this is how this constraint is expressed and which means that this fundamental matrix has a 0 vector and from the linear algebra it says that; that means, this fundamental matrix is a rank deficient fundamental matrix. So, if I take the determinant of this matrix you should get 0 so, that you can also check with this result.

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Now, we will proceed with the next example and in this case we will consider a more general scenario of projection matrices where, we would like to compute fundamental matrix epipolar lines. And also I think you can compute also the for epipolar line you need to compute a epipoles definitely and there are interesting concepts of end image points of P prime. So, let us consider let us discuss this solution.

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So, if you would like to compute the fundamental matrix that is between P and P'. So, P once again is no expressed as in this form [M p4], similarly this matrix is M'. So, first we

need to compute the homography at infinity what we discussed earlier; that means, we need to find out the homography between the corresponding points, between a point and the corresponding point of the same point which is lying at plane at infinity.

So, let us discuss these issue. So, we have a same point suppose this is plane at infinity. Now, for this convenience I am just showing it in the diagram as you understand plane at infinity is not realizable physically, but mathematically there exist a plane. And how a point in a plane at infinity will be represented? You consider any particular direction and then you use the scale factor 0. So, this is how the point obtain at infinities you know represented. So, this is my same point say X and in a stereo setup I would be considering the corresponding images of this point.

So, this is an image point x and this is so, this is my camera center. So, this image is x' and we are considering homography between these two and which we will be expressing as H infinity. So, how they are related with this let us see. So,

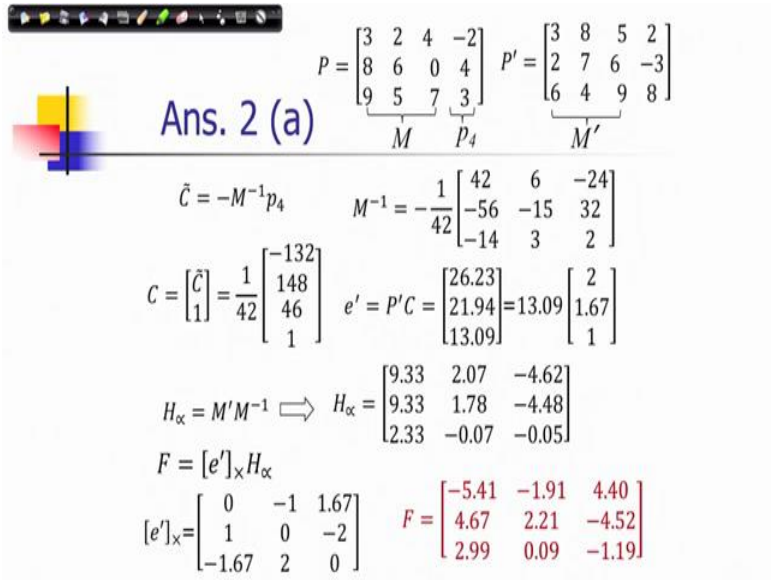
$$x = P\begin{bmatrix} d \\ 0 \end{bmatrix}, x' = P'\begin{bmatrix} d \\ 0 \end{bmatrix}$$

So, $x' = (M'M^{-1})Md$

You note that $M^{-1}M$ makes an identity matrix so, this will give you this. So, this can be considered as a 3 X 3 matrix H infinity and Md is x. So, you find there is a homography between these corresponding points. So, this is what we will be using here in this case because we know that fundamental matrix can be considered if I know the right epipole here.

See if the right epipole is e' then fundamental matrix is given by e'XH infinity that is what we discussed previously also. And then we have used the expansion of H infinity to get all those forms, but we will be using simply we can compute H infinity. So, in this problem what we will need to do, first we will be computing e'. So, e' is nothing, but image of the camera center of the first camera. So, we have to compute the first camera and then take the image which will give me e'. And, then we need to compute H infinity from M and M' by performing this operation which is $M'M^{-1}$ and then by performing this we can compute the fundamental matrix. So, let us carry on this computation as we discussed.

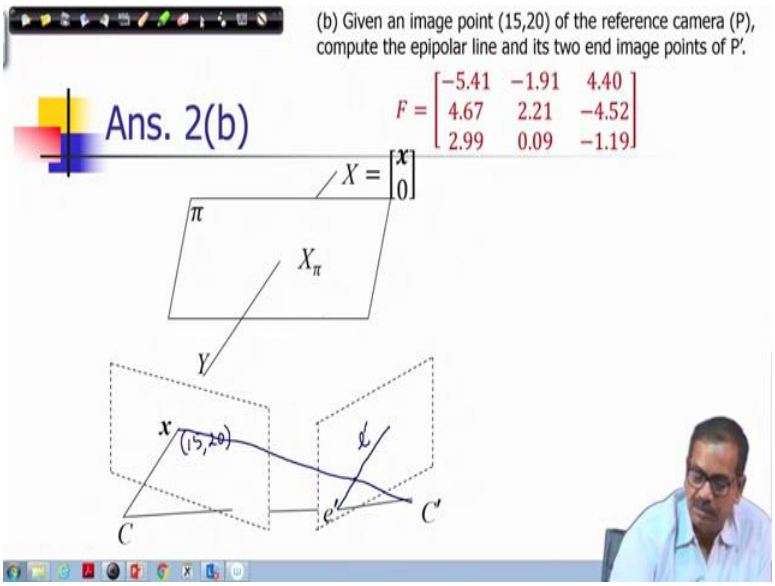
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So, we have computed the camera center in this form. So, we need to perform the M^{-1} operation and then now you can compute the camera center $\tilde{C} = -M^{-1}p_4$ which is given in this form in the projective space. And then compute the right epipole as P'C which is given in this form. This is the image of the camera center and you which is giving you the right epipole and this is right epipole. So now, you have to perform the $[e']_X$ you have and you should compute the homography at infinity as we discussed which is $M'M^{-1}$.

And if I perform those operations we will get this homography matrix. So, fundamental matrix is $[e']XH_\infty$ and which will give you this particular form. So, $[e']_X$ is this one so, if you find out this is e' I can get the $[e']_X$ I can represent. So, I am ignoring this scale factor here, scale value is not important here and then you get the answer as in this form. So, you note that fundamental matrix is also an element of a projective space. So, if I scale these values it will also denote the same fundamental matrix.

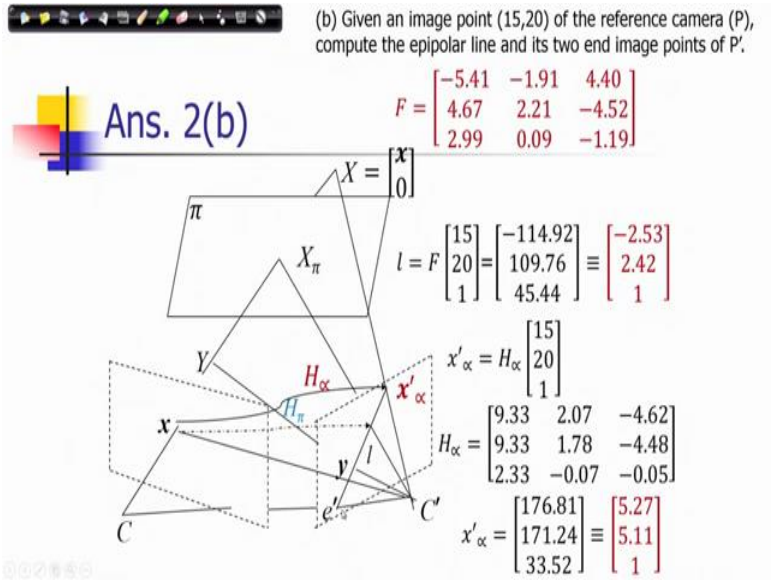
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So, the next problem is that if I give you an image point which is given as say [15 20] in the reference camera. So, in this case suppose this point is denoted by the coordinates of 15 20 then I need to compute the epipolar line. So, which means if I join this, so, I need to compute the epipolar line so, this line.

So, what I need to compute? I have to compute the corresponding, so this is a thing what I need to do. So, what I will do? I will multiply this point with F then I will get this epipolar line that is the relation that is how fundamental matrix is related with a image point to its epipolar line.

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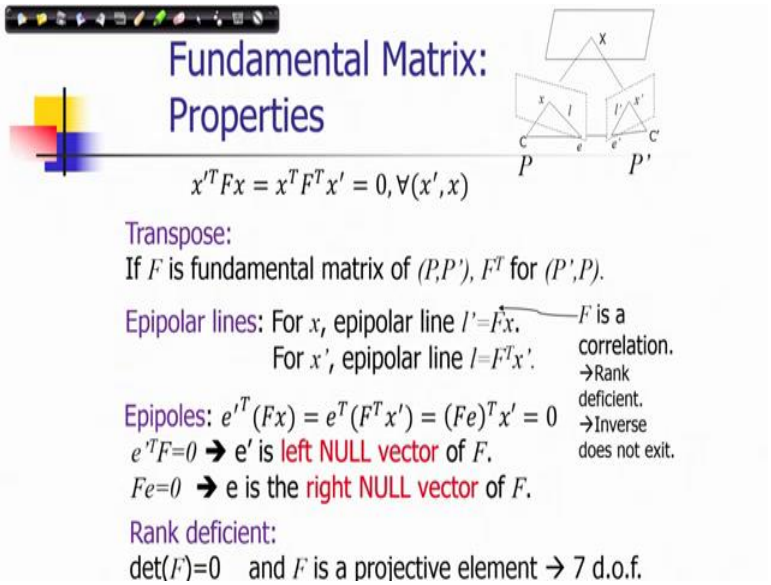
So, this is what we need to so, it is just showing that now all the points along these rays which is lying there. And also as we have discussed that there would be homography between these points, but finally, the point which is lying at infinity that is the limiting point. So, you see that is that concept of vanishing point is also here in this kind of epipolar line constraint. No other image point will exist, it will not there will not be any intersection beyond this point. So, you have an epipolar line, you have a very finite representation of epipolar line.

At one end there would be right epipole and the other end there will be the image point corresponding to the homography of plane at infinity or corresponding point of the same point which is lying at the plane at infinity. So, this is what we sometimes represent in this notation also and this is this diagram is explaining this part. So, this is how the epipolar line is computed, I multiplied the point [15 20]. The representation is [15 20 1] in the homogeneous coordinate space and if I multiply it with F. So, I will get the epipolar line in this form which I can again reduce in into this form by taking the third dimension scale equal to 1.

So, which means the equation of this line will be - 2.53x+2.42y+1=0 and to get the x infinity because, in this case problem also you wanted to know the end points of the epipolar line. We can compute or we have already computed homography at infinity simply you can multiply with this point and applying this we can get this is a point. So, we

know epipolar epipoles of in the right image plane and also the corresponding end point of that epipolar line in this way.

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So, now let me summarize the properties of fundamental matrix. So, we have discussed and we have also shown how you can compute fundamental matrix given the projection matrices. How you can compute the epipoles, again given the projection matrices and also we have discussed how the homographies are particularly homography at infinity is inducing induced in this case. They are they can be they are related how the how it is related also with the elements of projection matrices.

So, now let us consider the properties of fundamental matrix, we will be summarizing some of them we have already highlighted. So, the very basic property that if you get two corresponding image points of the same scene point then they are related by this particular relationship $x'^T Fx=0$. When we considered the reference image point is x and the second image point is x'.

If I consider the other way then again you can use we can also convert this relation as $x^T Fx'$ which means x' is the reference image point. So, you note that the matrices F and F transpose they are related in this fashion. So, F is the fundamental matrix of this configuration and F transpose also is a fundamental matrix of the stereo configuration, when we change the plane of reference or reference camera, image camera.

And this is true for any pair of corresponding image points that is the property first property. So, next is this transposition what you have observed that if F is the fundamental matrix of camera setup P, P'. So, we will be denoting a stereo setup by a pair of camera matrices. The first one in that couplet the first one we will denote, the camera matrix of the reference camera.

So, P, P' if F is the fundamental matrix for that setup and for P' P then F^T will be its fundamental matrix. Then the properties with epipolar lines, that if we have an image point x then epipolar line in the second camera is expressed as l'=F x. Similarly, if you have an image point in the other plane x' its epipolar line in the reference plane would be $l=F^Tx'$.

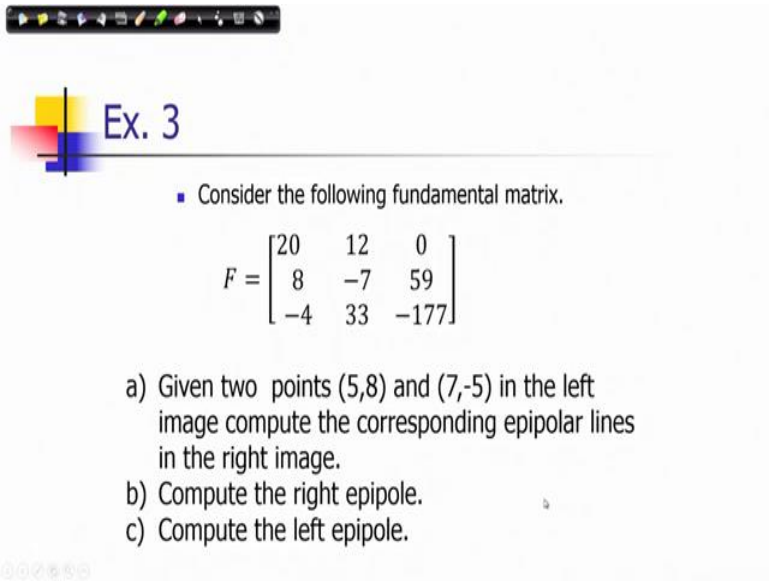
And this is what I mentioned earlier that, if this relation is called correlation because it is converting a point into a line, its transforming this kind of transformation is known as correlation. F is also rank deficient we discussed this while discussing about the solution of a problem. And so, which means that its inverse does not exist and how then this epipoles are related with x that we discussed. In fact, epipole also they are lying on the epipolar lines always they will lie on an epipolar line. And they are the intersections of all the epipolar lines which is given by this e prime transpose F x that should be equal to 0 or e transpose so, these are the conditions.

So, this should be equal to 0 or $e^TF^Tx'=0$, $Fe^Tx'=0$. So, it is just denoting the point contentment relationship of epipoles on their respective epipolar lines. The other thing what we I have discussed just a we made a very brief mention about that, that F e equals 0 which means e is the right null vector of F. Similarly, e prime transpose F is also equal to 0 and e' is the left null vector of F in the same way. So, these are some important properties of fundamental matrix and this is another property that determinant of F is 0.

Because, it is rank deficient we have mentioned we have discussed that and particularly it is interesting to note that how many independent parameters are there. So, one thing I already mentioned that there is a scale factor associated with F which means if I scale the elements still it will give me the same, it will express all those relationships; you can note from the relationship itself. So, for example, you take the first fundamental relationship between the corresponding points say $x'^TF\ x = 0$, if I multiply F with an scalar value k still this relationship holds.

So, you can check with any other relation that would be the case and then this so, this is one particular constraint that the scale is there is a scale factor. So, which means F has a 9 elements and it reduces one particular parameter by that constraint. The other thing is that its a rank deficient, its determinant is 0 that is the second constraint. So, there will be 7 independent parameters out of those 9 elements so, its degree of freedom is 7.

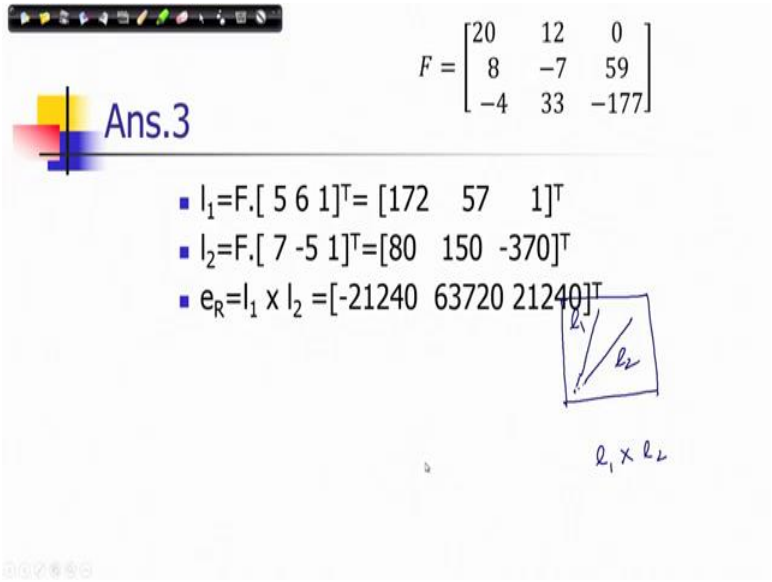
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So, we will be using this property to discuss about know to you know solve a problem here. Suppose, you are given only fundamental matrix, in our previous problems we have given you the projection matrices and from there you could very easily compute the epipoles. Because, by applying that property that epipoles are images of camera centers and given projection matrices you know how to compute camera centers. But, now if I just give you simply fundamental matrix; can you compute its epipoles?

So, this is what we will we would like to say, first thing is that in this we are also given an additional problem that is to compute the epipolar line. So, the first problem is about computation of epipolar lines. Suppose you have two image points 5 8 and 7 - 5 in the left image and you have to compute the corresponding epipolar lines in the right image. And then you compute the right epipole and compute the left epipoles. So, as I suggested previously also you should stop, you should give a pause at that in my video and then again you should work out on this problem. So, let me discuss the solution of this problem.

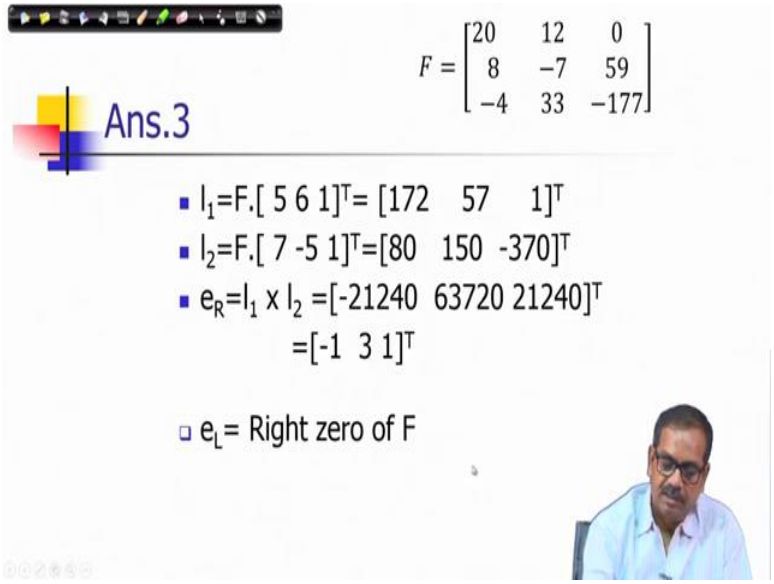
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So, consider this is the fundamental matrix and for the image point [5 6] its epipolar line is given in this form. If I multiply with F, I will get its epipolar line l1 which is given by this vector, similarly I can get the epipolar line for the other image points [7 - 5 1]. So, you have two epipolar line so, that is giving me the answer for the first part. But, one interesting thing I should note here you should note here that suppose this is your image plane where epipolar lines are computed. That means, is second cameras, image plane of a second camera; say this is your l1 and say this is your l2.

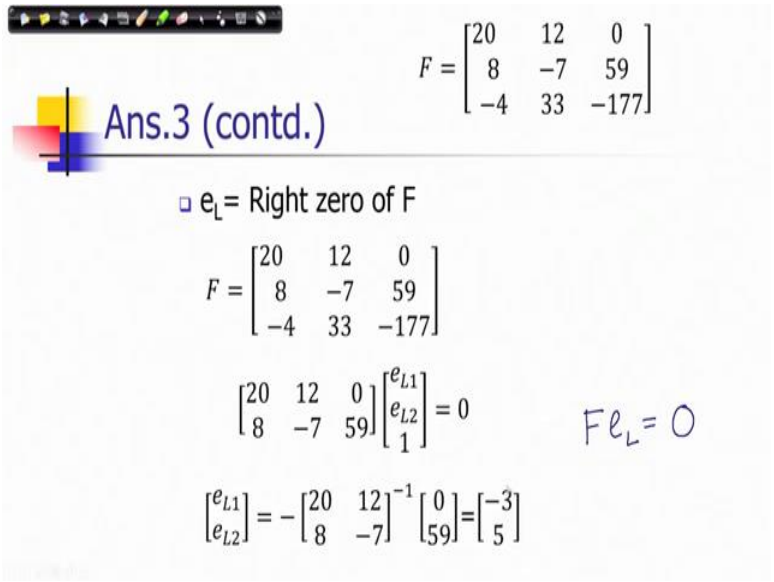
So, you know that all epipolar lines they intersect at epipoles. So, simply by finding out the intersection of l 1 and l 2 I can get the right epipole. So, what I need to do? I need to perform only l 1 X l 2. So, that is what we can do here. So, if I perform that l 1 X l 2 I will get a epipolar I will get the epipole. I mean it is a big numbers, but you can always not choose the scale to make it a smaller value, make all those values small; I mean which is shown here as [- 1 3 1] that is the value.

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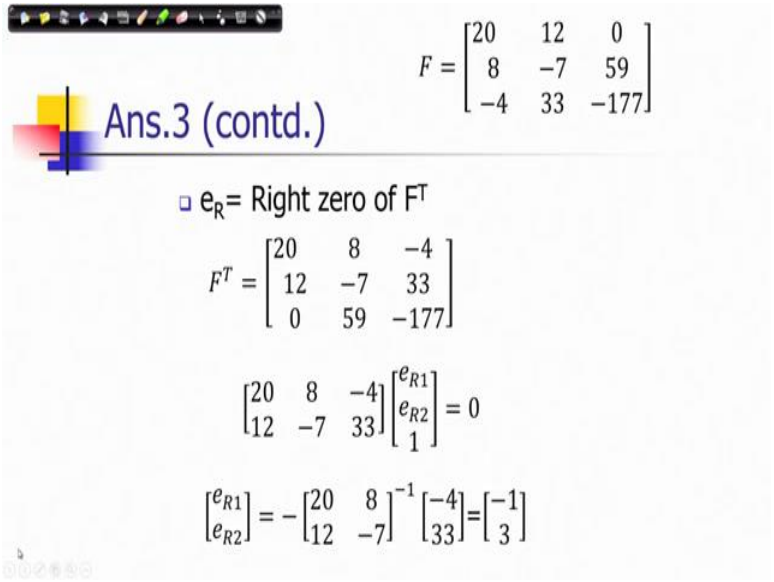
So, the coordinate of epipole in the right plane is [-1 3] in the conventional or two-dimensional real geometry. To compute the left epipole I could have taken note to such any two arbitrary image points in the right image plane and find out they are epipolar lines and apply the same technique. But, instead I will be discussing about right zero by finding right zero of F because that would also give me the left epipole which means I have to compute F.

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The right zero means that if I multiply the left epipole with F, it should give me a 0 vector which is a know 3 vector form. So, this is how this competition is shown here. Since, it is a ranked efficient matrix one of the row I can ignore, there are only two independent rows. And then I will convert this equation, I will assume that the coordinate is given of the right zero is given in this form [eL1 eL2 1]. So, I have to take the inverse of this part minus of this I can reduce this part, use the sub matrix operations and you can show that this is nothing, but computation of this particular configuration. So, you will get the right zero of F as [-3 5]; I mean it is giving you the coordinates already in the image plane, image coordinate plane, non-homogeneous coordinate system because scale factor you have already assumed as one.

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I will show you also that actually the right epipole what you have computed as an intersection of two epipolar lines we can compute it also a right zero of F^T. So, you take the F^T so, just transpose the matrix F and once again apply the similar technique; that means, ignore the third row and convert the equations, you form that these two equations or set of equations in this form. And then you solve for the corresponding right epipoles and you will get again you see that you are getting the same answer [-1 3]. So, with this let me stop this lecture at this point. We will continue this discussion in my next lecture.

Thank you for listening this lecture.

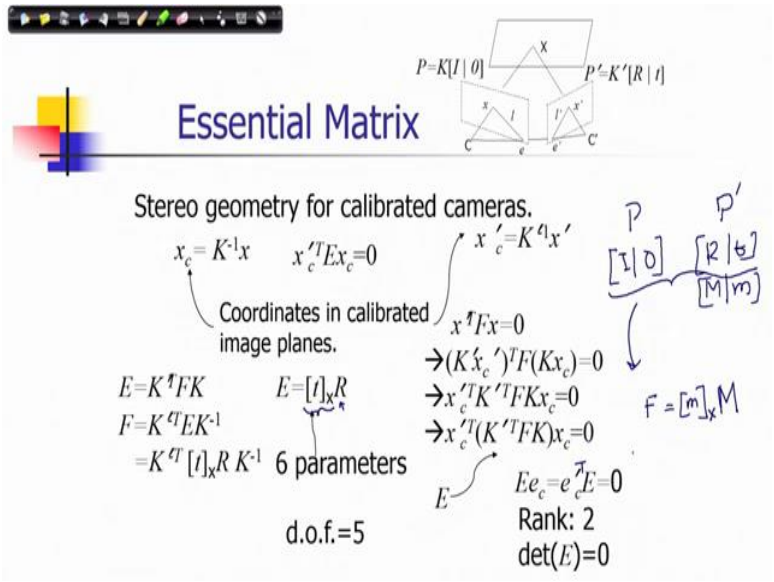
Keywords: Fundamental matrix, essential matrix, right epipole, left epipole

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Lecture -18
Stereo Geometry Part- III

We will continue the discussion on Stereo Geometry and we discussed the concepts related in this geometry like epipoles, epipolar lines, epipolar planes, then the fundamental matrix and how fundamental matrix converts an image point to an epipolar line in the other plane where the corresponding image point lies. We also discussed about the fundamental matrix of calibrated camera setup when this matrix is called essential matrix. Now in this lecture, we will further elaborate the parameters or elaborate the properties of this essential matrix.

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So, in brief, it can be called as stereo geometry for calibrated cameras. So, in this case, we represent say the projection matrix P=K[I|0]. So, since the calibration matrix I can convert it into a canonical form of [I|0] and also the other matrix I can also convert it in its canonical form, because just by applying rotation and translation parameter itself all the image coordinates are represented in normalized image coordinate system.

So, I can also convert it as [R|t]. So, this is how in the canonical form the coordinates can be converted just by multiplying with the inverse of the calibration matrix. So, you see that in the first camera x is converted to xc which is the coordinate in the canonical setup

optimal camera. Similarly x' is converted to xc' by applying there inverses of the respective calibration matrix. So, this is what is the expression.

So, now, if I apply the relationships of fundamental matrix in these setup, then that fundamental matrix is called essential matrix or E, but the relationship remains the same.

$$x_c'^T E x_c = 0$$

You note that in general it is $x'^T E x = 0$ when we do not have any idea of K and K' from the projection matrix. So, how an uncalibrated fundamental matrix; that means, fundamental matrix for uncalibrated stereo setup is related to a calibrated stereo setup? So, how they are related? So, their relationships can be expressed using those calibration matrices itself.

As you can see with this simple derivation that

$$x'^T F x = 0,$$

then convert x'^T into its canonical form as $K' x_c^T$. Similarly Kxc and then you consider the middle part of this composite matrix. So, that is the 3 X 3 matrix. So, this matrix is nothing, but this is the essential matrix E. So, this is what is E. So, the relationship between fundamental matrix and essential matrix can be expressed with using these calibration matrices as

$$E = K'^T F K$$

K' is the calibration matrix of the second camera and K is the calibration matrix of the first camera.

Similarly you can also get fundamental matrix from essential matrix by using those calibration matrix. So, the calibration matrix, it reduces into this from because if I consider once again you note that relationship; if you remember it. Suppose I have [I|0] and then [R|t] see these are the 2 camera matrices, this is P and this is P'. So, if you remember that this structure was [M | m] in general and the fundamental matrix was given as this $[m]_X$ matrix, which is a skew symmetric matrix into M. So, that is how the fundamental matrix can be derived.

So, in this particular setup using the same relationship I can write

$$E = [t]_X R$$

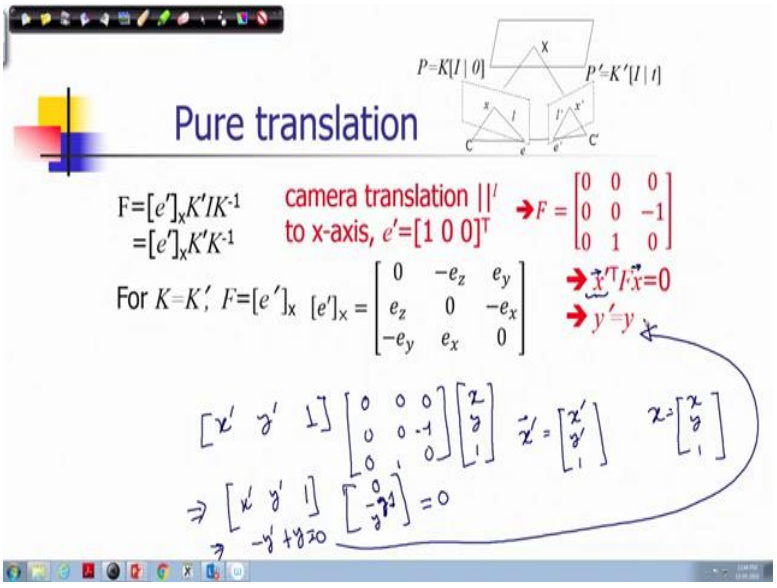
So, this is how the essential matrix can be represented or can we derive. So, I will replace that value. So, here you note that there are only 6 parameters involved in this particular operations like 3 parameters related to the translation of camera center. There is a translation parameter and rotation of the orientations rotation of the axis, so that is also 3.

So, there are 6 parameters which are involved in constructing the essential matrix and since scaled is involved; that means, if I multiply essential matrix with a scale value K still all this relation holds. So, out of them one again 5 are independent and also the epipoles in its canonical representation, so it also has those relationships holds; that means,

$$Ee_c = e_c'^T E = 0$$

There should be a transpose here, because it is a right zero, but we have to take it a transpose operation and its rank is also 2 like F. So, det(E)=0 and as I mention d.o.f of E=5, because a scale factor is associated with E and it has 6 parameter. So, it should be 5.

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Now, consider a situation of special kind of stereo set up where you have pure translation of camera centers, pure in the sense there is no rotation of axis. So, in that case we can very easily express the projection matrices having a simple structure like if you have pure

translation, then projection matrices can be expressed as say K[I|0]. I am representing again in the uncalibrated form then P'= K'[I|t]. So, simple translation; so, K't is that translation of the camera.

So, your fundamental matrix is expressed in this particular setup is in this form

$$F = [e']_X K' I K^{-1}$$

You remembered that if it is in general case it should be rotation matrix R. So, in this case rotation matrix itself is an identity matrix because there is no rotation and then this can be reduced as

$$F = [e']_X K' K^{-1}$$

That is a structure of fundamental matrix and if I consider both the calibration matrix is a same which means they are the same camera. They are using the same camera for taking the images or it has been fabricated in such way that all other intrinsic parameters they remain the same. So, then fundamental matrix takes a very simple form of $[e']_X$

So, just from the epipole you can compute the fundamental matrix itself. Whereas, $[e']_X$ is given in this form;

$$[e']_X = \begin{matrix} 0 & -e_z & e_y \\ e_z & 0 & -e_x \\ -e_y & e_x & 0 \end{matrix}$$

This is the definition of $[e']_X$ and a since you have if you consider a special kind of translation that it is a parallel to x axis then, e' is the vanishing point of x axis and which is given by [1 0 0]. So, only to nonzero elements and the structure is very simple. So, if I apply this relationship of corresponding points, then we can show that the y coordinates actually they remain the same.

So, in this representation there we need to highlight few things that x' here is a vector. So, x' is a vector and in my representation we consider for example, this points are say

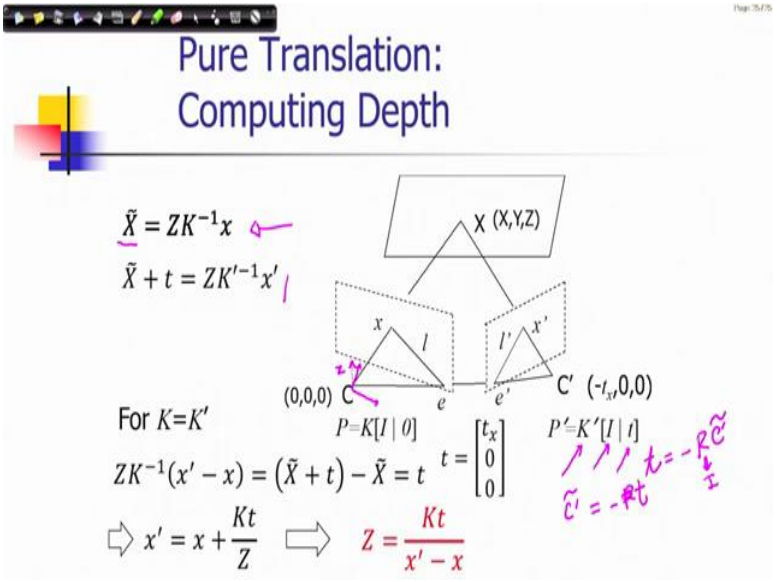
$$x' = \begin{matrix} x' \\ y' \\ 1 \end{matrix} \; and \; x = \begin{matrix} x \\ y \\ 1 \end{matrix}$$

$$F = \begin{matrix} 0 & 0 & 0 \\ 0 & 0 & -1 \\ 0 & 1 & 0 \end{matrix}$$

$$x'^T F x = 0$$

$$[x' y' 1] \begin{bmatrix} 0 \\ -y' \\ y \end{bmatrix} = 0 \rightarrow -y' + y = 0$$

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Let us discuss how we can compute depth under pure translation. Consider these two cameras in the stereo setup as we have already mentioned that we have the corresponding camera matrices P and P' and where P is given by the K[I|0] which means the center of the first camera would be at in the origin and also this translation between these two cameras, it is given by $t = \begin{matrix} t_x \\ 0 \\ 0 \end{matrix}$

So, you are considering that there is also no rotation. So, it means the horizontal direction it is a pure translation and there is no rotation matrix R. As we can see in the case of second camera matrix here, we have same identity sub matrix here in this part instead of a rotation matrix of or we can say rotation equal to the identity matrix which means no rotation and this is a translation part and this is a second camera whose calibration matrix is also different which is K' and our objective is that to compute the depth here.

So, in this case since it is a canonical form so, your principal plane is also the xy plane of the coordinate system in the first cameras camera centric coordinate system. So, this is your canonical form. So, this is the z directions if I consider. So, your depth will be the z coordinate in this case. So, let us see how we can compute this z considering the corresponding points x and x' in the stereo view. Now the center camera center of the second camera, it should be $(-t_x, 0,0)$. As you can see that translation t in this form we will give this particular form because as we know that the camera center c'. I am using tilde to denote that it should be expressed in word coordinate.

$$t = -R\tilde{C}$$

R = I. So,$\tilde{C} = -t$. So, which will give you (-tx, 0, 0) that is the camera center in this form in this kind of configuration and then we can compute the corresponding world coordinate of the same point. So, here same point is X and in our convention when I am using the world coordinate in nonhomogeneous coordinate convention I will be using tilde on top of it.

So, as you can see that what we have done here.

$$\tilde{X} = ZK^{-1}x \rightarrow \tilde{X} + t = ZK'^{-1}x'$$

Similarly if I consider the other camera, there also we can use the same convention, but since there is a translation involved here.

It is because here also there is no rotation. So, your depth your principal plane is also the same xy plane and you can use Z coordinate as the corresponding depth. So, now, if I consider K=K'; that means, two camera have the same calibration matrix then this relationship can be further simplified. It takes a very simple form and as we can see from here itself if we subtract this equation, if you subtract see this equation from this one then you will get this relationship; that means, you get

$$ZK^{-1}(x' - x) = \left(\tilde{X} + t\right) - \tilde{X} = t \rightarrow x' = x + \frac{Kt}{Z}$$

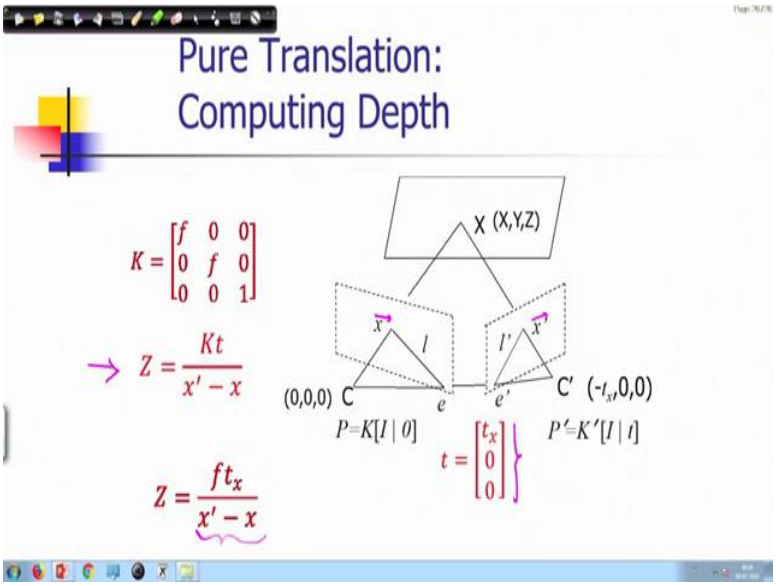
That is how we will get and from here we can get x'; that means, the second image coordinate of the since sin point in the second camera. So, this is a relationship between x

and x'. So, there is a shift in the horizontal direction and the amount of shift is determined given by this Kt/Z and from this shift you can determine the depth

$$Z = \frac{Kt}{x' - x}$$

So, this is how we can compute depth and this relationship can be further simplified once we take the take a simple form of K.

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So, in the calibration matrix we consider only focal length is a parameter and all other parameters they are initialized to 0, which means principal point is also the center of the image coordinates there is no skew and under this situation you have a very simple calibration matrix. So, this relationships; that means,

$$Z = \frac{Kt}{x' - x} \text{ and } t = \begin{matrix} t_x \\ 0 \\ 0 \end{matrix}$$

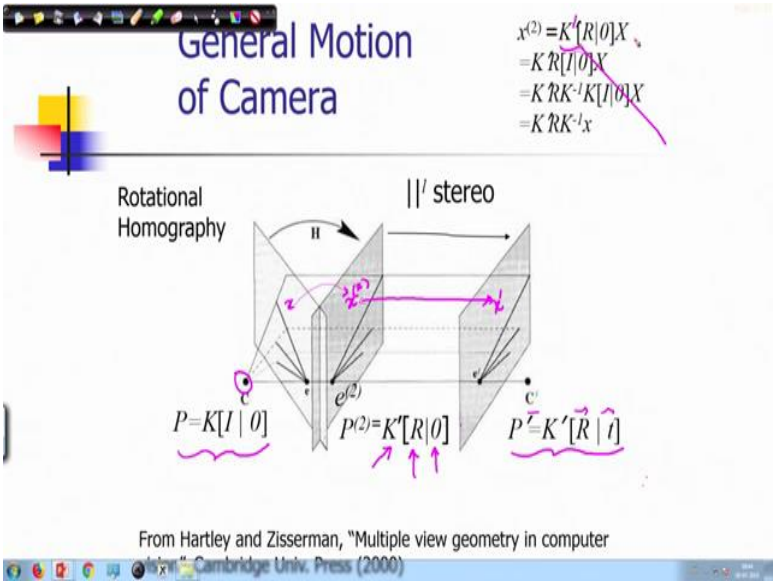
$$Z = \frac{ft_x}{x' - x}$$

So, this is one of the familiar equations of computing depth in a stereo setup where optical axis is parallel between two cameras and also both the camera have the same focal length. They have same identical calibration matrix and the shift between these 2 camera is tx then we compute depth incidentally this value x'-x which is a shift of the corresponding point

along horizontal direction or this is the x coordinates. These are all scalared x coordinates. You should note here once again that here this x and this x' it denotes the vector whereas, finally, in this computation when you have considered x'-x this is the scalar x coordinate of those two points.

So, this is also called disparity. So, in a stereo setup we can compute the shift or disparity and f and tx could be the parameter of the stereo imaging system. So, only using image, computations over those stereo images you need to compute disparity at every point and that is how you can obtain the depth.

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This could be further generalized when we have a general configuration or arbitrary configuration of two cameras instead of having a parallel stereo configuration. Let us consider that you have a general projection matrix for the second camera, whereas the first camera is a reference camera and there we follow the same canonical structure with the calibration matrix K and in the second camera we have these general structure where you have the calibration matrix K prime which is a different camera. It could have different focal length etcetera and it has rotation and translation parameters as well, which means you know they are not separated by simple horizontal motions C' or the center of the camera is rotated and translated with respect to C. It has a general configurations.

So, under this situation, how we can compute this depth that we would like to discuss. Now there is a simple way by which you can map this problem to a parallel stereo problem

and use the previous relationships what we derived here itself. So, what we can do? We can consider that let there be another imaginary camera which has the same center C, but it is related by only the rotation with the same rotation matrix R and then we can compute the homography between these two image planes. We know that it establishes a homography we have already discussed in two dimensional persecutory projection during projective transformation also and we will also see how this homography also could be, could be octant we can also find out here itself.

So, once we have these things for example, he you consider originally you have two point say x this is the reference camera. So, we will be considering this is x and another point say x'. So, these are the say 2 corresponding point of the same. Now due to rotational homography we bring x to the new imaginary camera setup where will be we will be considering this point is a x 2. Now these 2 since we have exactly applied the same rotational matrix R,

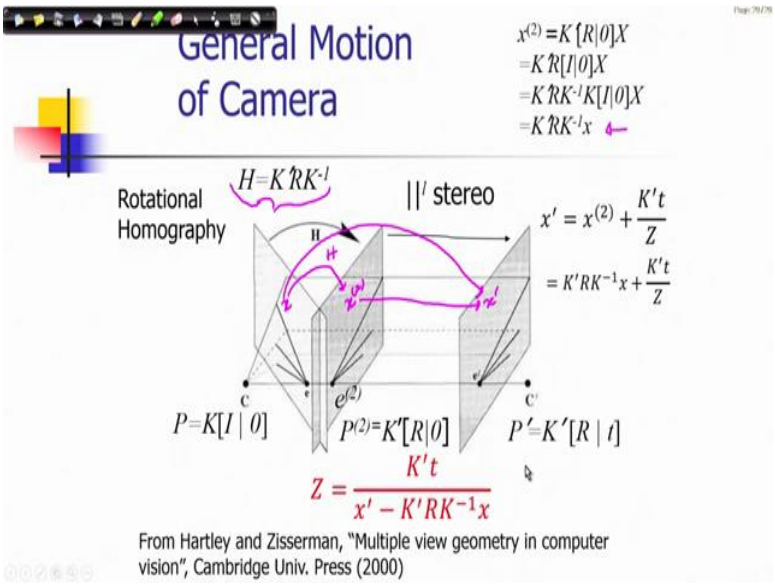
Now this image plane and this image plane they have they are they have that parallel stereo configuration. So, now, you can apply the parallel stereo results under the same; that means, what we need to do, we need to consider now the corresponding points under this 2 camera setups and x' and use them use this corresponding pair of corresponding points to compute the depth. So, let us elaborate this computations further. So, let us first compute the rotational homography in this case.

So assume that you have the second camera let me rub this thing just to make it more clear. So, consider that the second camera matrix as P 2 that is a imaginary camera as I mentioned which we considered whose image plane should be parallel to the image plane P prime after performing this rotational transformations on the image planes. So, P 2 since there is it is related with the original reference camera by the rotational matrix R and also. I would like we would like to have the same calibration matrix of the second camera K prime.

So, the projection matrix of P 2 is given in this way that it that it has a calibration matrix K' and in parameter of rotational matrix R and the center is still at origin 0 and that is the same as the reference camera. So, under this configuration what should be the homography between this 2 sins. So, here we can see that x2 can be written as in this case it is the image of the same scene point x. So, this could be written as K'. So, this is K' which is not very

invisible from this part. So, K'[R|0]. So, this projection matrix P2x that is what is x 2 further. Further simplifying it so, this is x 2 and once again we can simplify it in this form which means I can take R outside.

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So, it becomes K'R[I|0]X. Now this part can be expanded into this way. So, our objective is that out of this will be taking this part here and then K[I|0]X can be considered as x that is the original scene point. So, once again this is x2 which is mapped by homography H and then x'. So, originally x and x' is the corresponding points and now it has become x2 and x' is the corresponding point. So, now, the relationship between x and x2 and x is given in this form.

$$x^{(2)} = K'RK^{-1}x$$

So, the rotational homography under this situation it will become $K'RK^{-1}$. So, this is a rotational homography. So, these are rotational homography. So, once you have obtained this now you can apply the stereo results between x2 and x', which means that I can write

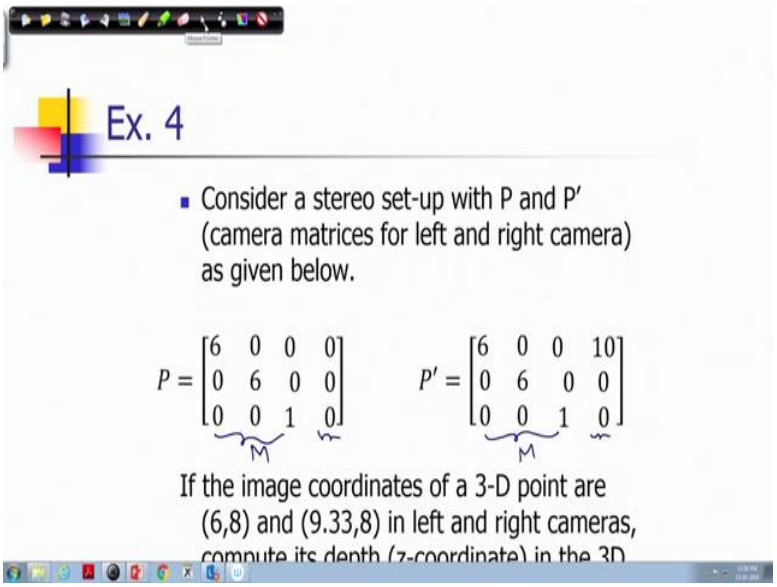
$$x' = x^{(2)} + \frac{K't}{Z} = K'RK^{-1}x + \frac{K't}{Z}$$

So, that is how x' and x is related. Now it is a simple mathematical manipulations algebraic manipulations.

$$Z = \frac{K't}{x' - K'RK^{-1}x}$$

So, from here we can get this relationship. So, in this way we can obtain the depth or Z coordinate under this situation of the same of this kind of arbitrary general motion of camera.

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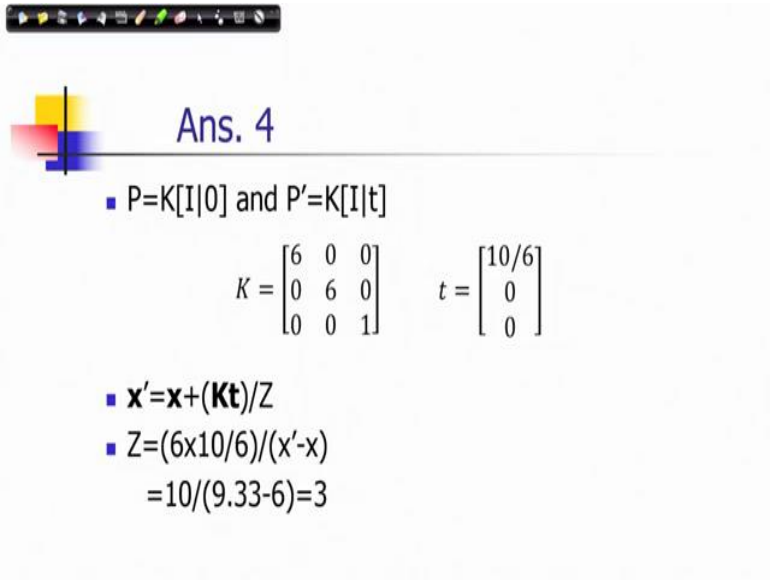
So, let us work out a problem to understand this relationships that how the depth could be computed or Z coordinate would be computed. You consider a stereo setup with these projection matrices P and P' which are given here as it is shown and now what you need to find out that you have been given 2 image coordinates 3-D points are there (6, 8) and (9.33, 8) in left and right cameras.

So, you should note that we have already satisfied that constraint of keeping the y coordinate same because that is what we discussed as a property of this particular scenario you can find out that the component of sub matrix m, they remain the same and only these column vector they are different. So, which means it is a case of camera cameras which are related with only translations of origins because you know that this is given by minus m inverse this column vector this is given by m inverse this column vector and that is a only thing because others are either rotations or rotations except there is no rotation relative rotation between this two system.

So, this is a case of your translation. So, in the under this case you we already seen that your y coordinates of the corresponding point they should remain the same because the there is a special case of its a special case of you know stereo and the fundamental matrix is given by simply by E prime cross as we have seen earlier also.

So now let us compute the depth of this particular you know in the particular what should be the depth of sin point?

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So, this is the scenario your P = K[I|0] and P' = K [I|t] where

$$K = \begin{matrix} 6 & 0 & 0 \\ 0 & 6 & 0 \\ 0 & 0 & 1 \end{matrix} \; and \; t = \begin{matrix} 10/6 \\ 0 \\ 0 \end{matrix}$$

So, this is what is K and this is this is what is t. So, you note that t needs to be scaled by the value 6. Because in the previous case it was Kt. So, you will get if I multiply this one then only you will get the previous matrix. So,

$$x' = x + \frac{Kt}{Z} \rightarrow Z = \frac{\frac{6 * 10}{6}}{x' - x} = \frac{10}{9.33 - 6} = 3$$

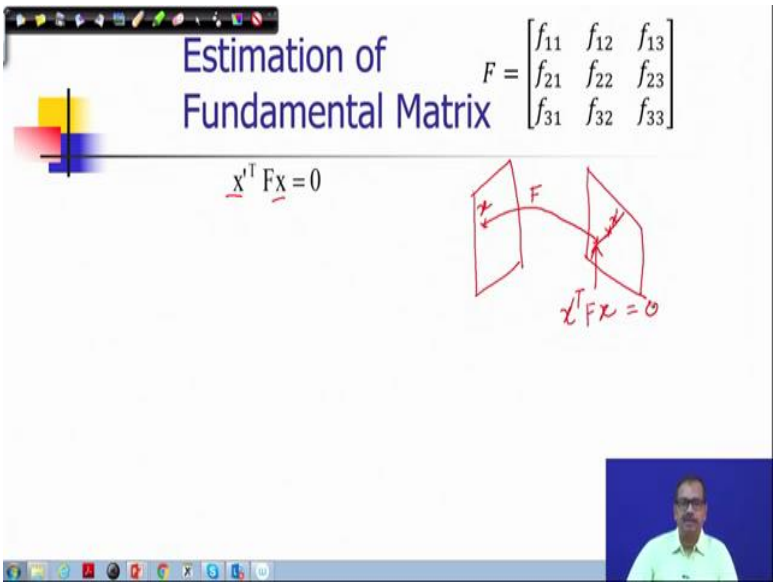
So, with this let me stop my lecture here and we will continue this discussion in my next lecture. Keywords: Essential matrix, Fundamental matrix, Projection matrix, computation of depth

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Lecture - 19
Stereo Geometry Part – IV

We are discussing about Stereo Geometry and we discussed how a fundamental matrix of this geometry characterizes the system. The in this lecture we will consider how to estimate a fundamental matrix given a pair of images of a stereo imaging system. So, you consider the relationships with the corresponding pairs of points for a fundamental matrix.

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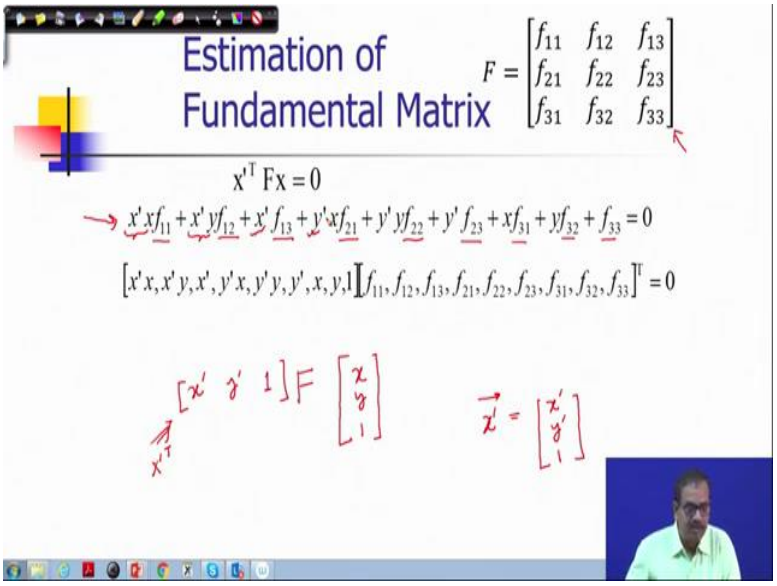
And as we have already discussed that you have two pair of points that is x'; that is in the right stereo image and x in the left which is a reference stereo image. Then the relationships with the fundamental matrix between and also this pair of points can be described in this form; that is $x'^T Fx = 0$; if I consider two images of a stereo system as I mentioned. And consider a point this is x; this is the reference camera system or reference image plane of the stereo system and its corresponding point x'.

Now, the task of a fundamental matrix or the property of a fundamental matrix is that; it transforms a point into the epipolar line. So, it transforms a point; if I apply Fx, it will transform the line where x' will occur. So, this line is given by this equation Fx and the

applying the point content print relationship of projective space; since x' lies on these lines; so $x'^{T}Fx = 0$; so that is how we have derived this relationship.

So now we will see that using this property how we can estimate a fundamental matrix given a pair of image points; so for that we need to know say a few corresponding pairs of points, so this is a problem now.

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So, this is the expansion of this particular equation. If I consider the element of the fundamental matrix is given in this form; it is a matrix element and using the notation of index indexing notations of an element of a matrix that element of an ith row and jth column is given by fij.

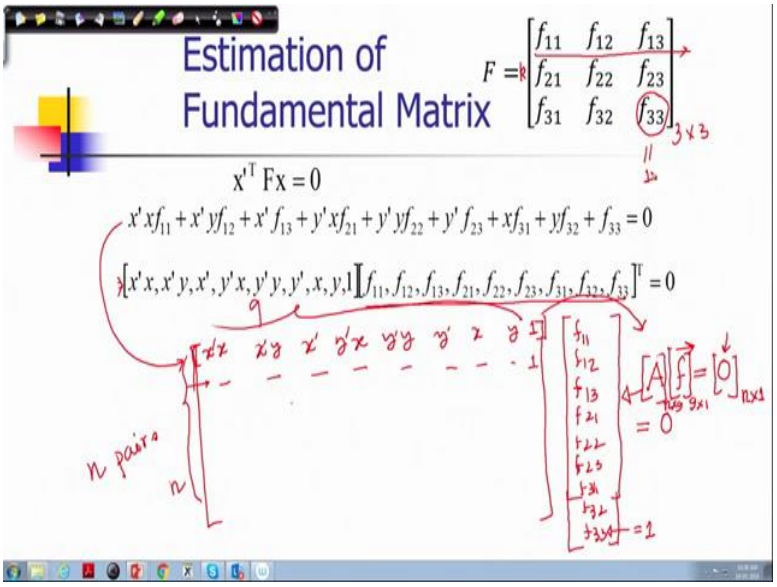
$$F = \begin{matrix} f_{11} & f_{12} & f_{13} \\ f_{21} & f_{23} & f_{23} \\ f_{31} & f_{32} & f_{33} \end{matrix}$$

And also you consider the column vector notation of a point for example, x' = [x' y' 1]. In our convention we consider a point x', but its x coordinate is x also denoted by this symbol x'; only thing is that it is scalar quantity of an x coordinate, then y coordinate and 1. So, this is the presentation of the point x' there. So, if I we can write it in this form; [x' y' 1], this is what is x' transpose. And then multiply with fundamental matrix F and also the point x in the reference image corresponding point which is also given by this column vector [x y 1].

So, now you consider multiplications of these matrices; if you expand, you will get this equation; And again it would be convenient for us to represent in the form of a matrix because in that case a set of linear equations can be represented by the matrix operations. So, a single equation is given in this way and in the matrix notation, here our objective is to compute the fundament elements of fundamental matrix; so, these are unknown variables or unknown parameters in our case.

And given the observation these values we already because that is what is given to you; I mean we will discuss later on how we can also compute these corresponding pairs in my next topic. But for the time being; let us assume that these points are given to you, maybe even through manual observation in a crude way you can get a few such matching points. So, these values are given to you; that means, these coordinates are given to you. So, now this equation in the matrix form is written here.

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Let me write it in a proper notation which is more convenient to understand. So, we will see that these are the coefficients of those equations.

$$x'xf_{11} + x'yf_{12} + x'f_{13} + y'xf_{21} + y'yf_{22} + y'f_{23} + xf_{31} + yf_{32} + f_{33} = 0$$

So, I am converting this equation in the matrix form which is written here also. So, because this is for the sake of convenience a column vector in this slide is shown by a row vector by using transpose operation; so, you are computing all these elements of fundamental

matrix. So, you can see that all the rows are again represented as a column vector and they are; they are concatenated in this form.

$$[A]_{nX9}\left[\vec{f}\right]_{9X1} = [0]_{nX1}$$

So, this equation is formed with only one curve pair of corresponding points; as I explained.

Suppose you know a few more points so; that means, I can have; I can represent all these equations in a general matrix form. So, all the coefficients of those equations will come in this matrix. So, this is for the first pair of points; for the next pair of points once again I will put these elements observations and I can write this equation. So, in this way if I know n pair of points; n pairs of points each one will give me one equation. So, as you can see what is the number of columns in this case? You have here 9 unknowns because fundamental matrix the dimension is 3 X 3; so, there are 9 unknowns. So, in this form we have I mean there are 9 columns and number of rows would be n and there are 9 columns.

So, this vector can be shown as a column vector f and this matrix let me write it as an A. So, $[A]_{nX9}\left[\vec{f}\right]_{9X1} = [0]_{nX1}$; so 0 is also a column vector because each equation is giving me 0. So, if there are n equations; so 0 will be a column vector of dimension n X 1 and as we know this should be 9 X 1 and this dimension should be n X 9. So, given n pairs of points corresponding points; you can form these set of equations.

So, now your task would be to solve this equation to get f and you can see this is a homogeneous equation; a set of homogeneous equations, again you can apply the least square error estimate to solve this; however, you can make this equation as a non homogeneous set. Suppose I consider the; I said this f33=1; like we did earlier also. So, one of the value let us consider say to certain particular value because I; I mentioned that though there are 9 elements; number of independent parameters in a fundamental matrix is 7.

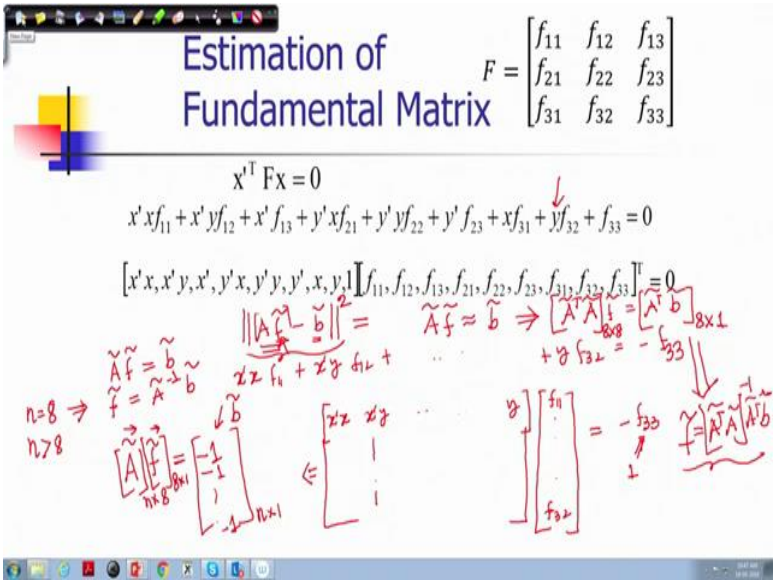
First thing that it is an element of a projective space. So, if I multiply all the elements by is constant k still this relationships it holds; that means, that is also a solution of this system. So, fundamental matrix multiplied by any scalar constant they are all equivalent that we discussed earlier. So, one of the element could be considered as a scale factor say if I take

this equal to 1; this would be taken care of. So, in proportion to that value all other values are expressed that is one particular feature property; so, that it is at least one reduction of the number of independent parameters.

The other constraint in a fundamental matrix that we also discussed that it is a singular matrix. So, if I take the determinant of this matrix; it will be 0, so you will get another equation. So, if I expand determinants you will get equations in terms of all these elements and that should be equal to 0; so that is another constraint. So, that is why there are number of independent parameter is 7.

Now, for the time being let us consider that now we will be solving for 8 parameters. We will later on we will see that how that properties of singularity could be enforced; for the time being let us let us approximately estimate fundamental matrix where the singularity constraint is waived. So, in that case as I mentioned that one of the parameter you can set to some known value; some given value. So, let us check to this value 1 and then you can convert this equation in this form. So, let me rub this part.

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So, once again; I can write this equation as

$$[x'x, x'y, x', y'x, y'y, y', x, y, 1][f_{11}, f_{12}, f_{31}, f_{13}, f_{22}, f_{23}, f_{31}, f_{32}, f_{33}]^T = 0$$

so this is a transformational. So, now with this equation becomes non homogeneous equation because this part is not 0. And in this way same for all the n corresponding pairs; you can get n such equations.

So, let me consider that matrix say A' matrix. So, A' into let me use also f prime; these are the dimensionally reduced matrices because of this conversion that should be equal to minus f 33. For example, if I set minus if set f 33 equal to 1; so this is the equation we will get ah; actually you will get a column vector of minus 1 because every question will have minus 1.

So, the dimension of this should be n cross 8; for dimension of this vector would be 8 cross 1 because there are 8 unknowns and this is n cross 1. So, applying the solution for system of non homogeneous equations; you can perform the least square estimate and you know that. But if I consider this column vector as say b; so I would be solving

$$\tilde{A}\tilde{f} = \tilde{b}$$

And since there are more number of equations and usually; you should have more number of equations at least minimally you require 8 equations to solve it then exactly you get a solution. Because this matrix would be 8 X 8 and if it is not a singular matrix you perform $\tilde{f} = \tilde{A}^{-1}\tilde{b}$ that would give you; so, when just you have 8 corresponding pairs of points I can simply write $\tilde{f} = \tilde{A}^{-1}\tilde{b}$

So, that would be the solution in that case and that is a requirement you should have 8 numbers of corresponding points that is the minimum requirement. And when it is n greater than 8; then what you need to do? You need to perform a least square estimate; that means, you will have to minimize the error.

So, error is defined in this way

$$\left|\left|Af - b\right|\right|^2 = 0$$

so the deviation of the vector b from vector Af. These dimension of Af would be also n X 1 and b would be n X 1. So, if I take the norm of this vector which is nothing, but the sum of the square of its elements and you can take the square root also or sum of the square of

its elements. And that should be I mean you can use also square just to show that square of the norm; that should be equal to that you have to minimize this.

So, you have to find out that f which will minimize this norm, can be solved by least square error method. There is a nice derivation we can perform by applying almost like matrix operations. But we should use the theory of minimization in this case by taking the derivatives of these expression and equating with 0; we have to solve these equations then you will get these solutions.

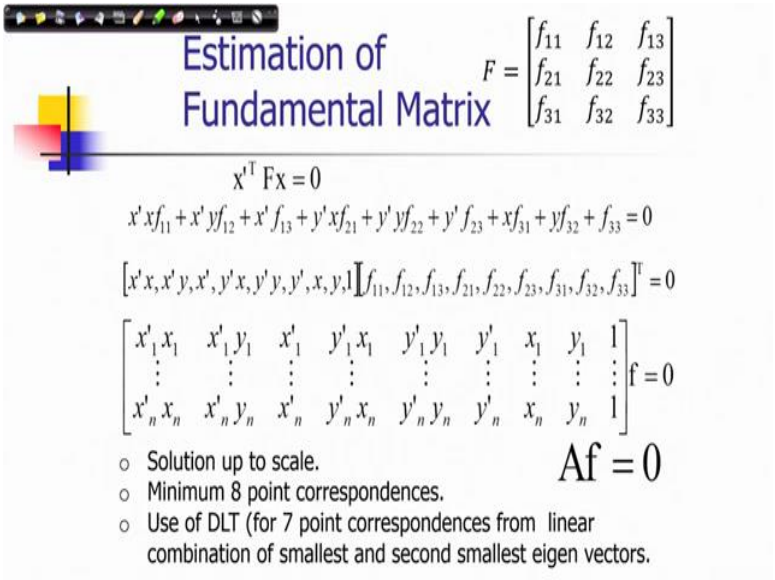
Just for the sake of your easy this aspect

$$\tilde{f} = [\widetilde{A^T}\tilde{A}]^{-1}\widetilde{A^T}b$$

So, now you see that this becomes a square matrix. So, this is this square matrix is in the form of 8 X 8; A is n X 8; so a transpose is 8 X n into n X 8. And of course, the whole thing this would be 8 X 1.

So, this is your least square error solution that is for the set of non homogeneous equations. So, let me summarize these expressions because I have also the slide set for this representation.

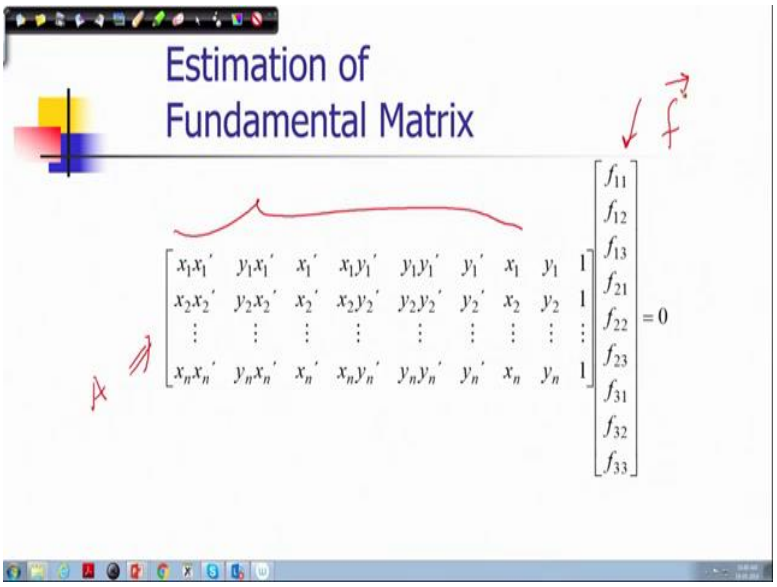
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So, as I mentioned this is how we represent the system of equations given n pairs of corresponding points. And in our discussion we also mentioned that the elements of f; they

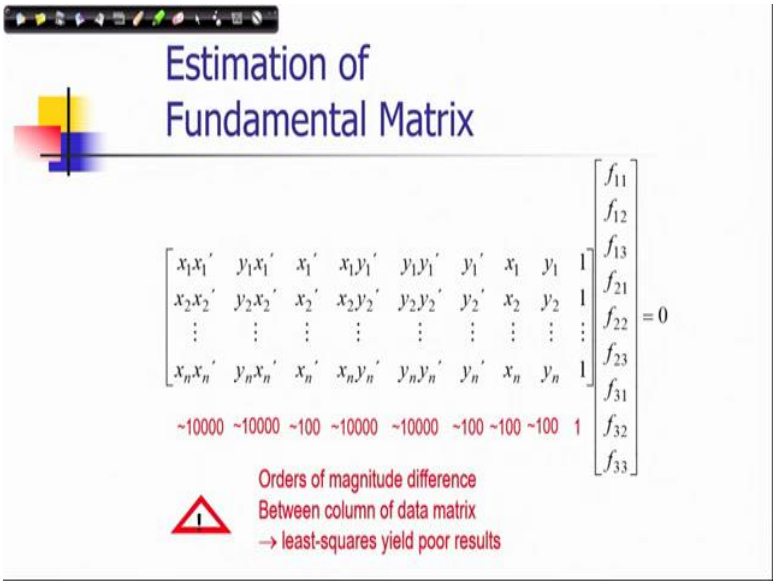
can be retrieved up to scale; that means, whatever solution you get if you multiply by any scalar constant; that would be also a solution of this equations. And also we discussed how minimum 8 point correspondences are required. There is a technique which we will discuss later on that even you can do its 7 point correspondences; just to mention that because as I mentioned there are 7 independent parameters. So, we will discuss this technique in my next slides.

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So, this is a representation of the set of equations with n pairs of corresponding points and you can see this is that matrix A; I was referring at. So, this is matrix A and this is the f that is a column vector of consisting of elements of fundamental matrix; that is the unknown and this is a set of equations.

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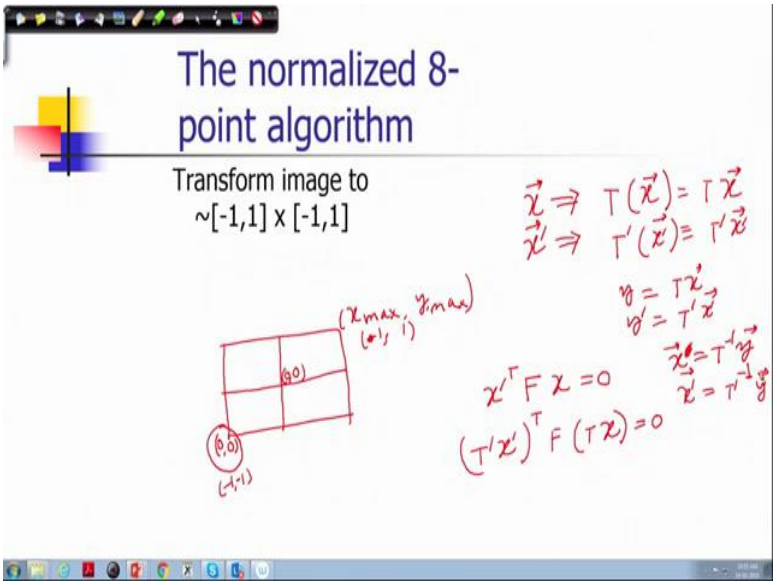


One of the problem with these presentation is that the coordinates of these points; they have a very large range. And as you are taking products in some cases; you understand product of two large numbers will be also very large; so, the dynamic range of these coefficients of this equation of the unknown parameters; that widely vary.

You can see that for the products; the ranges could be in the tune of minus in sorry in the tune of 10000; this could be magnitudes that is the variation ranges. It could be negative positive whatever because the coordinates could be usually in the images all the coordinates are positive; so, let us assume all values are positive. So, so this is the range 10000; 10000 and whereas, the coefficients with single variables range would be smaller 100; assuming the images dimensions are in that 100's only. So, 100 X 100 image, but then you get all the; you know coordinates within 100.

So, that is why since the orders of magnitude to difference it is; there is a lot of difference in that orders of magnitude. So, numerical stability is less in these computations and it yields poor results. So, we will see how to handle that part let us proceed.

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So, one of the technique to take care of this particular fact that why not you transform the coordinates within a smaller; within this unit range of [-1, 1], then you can see even the products of coordinates also they have equivalent equal ranges of the range of the single coordinates; all the coefficients they will have this.

So, one of the technique is that you can transform these coordinates between [-1, 1] X [-1, 1]. That means, suppose you have your image coordinates it varies; if I assume say this is 0, 0 and say this is the width and height of the image say (xmax, ymax). So, these are the coordinates within this ranges all the coordinates are represented. So, now we I would like to transform this coordinate.

So, the variations are there; that means, exactly in the middle you have the centre point is represented by origin (0, 0); so this is how coordinate is transformed. And you can see that now if with transformations you get transformations in both the images independently, still you can recover the fundamental matrix that also let us understand. Say you have an image as I mentioned say any point x; now it is transformed to a point, this transformation is given by this in the say left image; that means, a reference image.

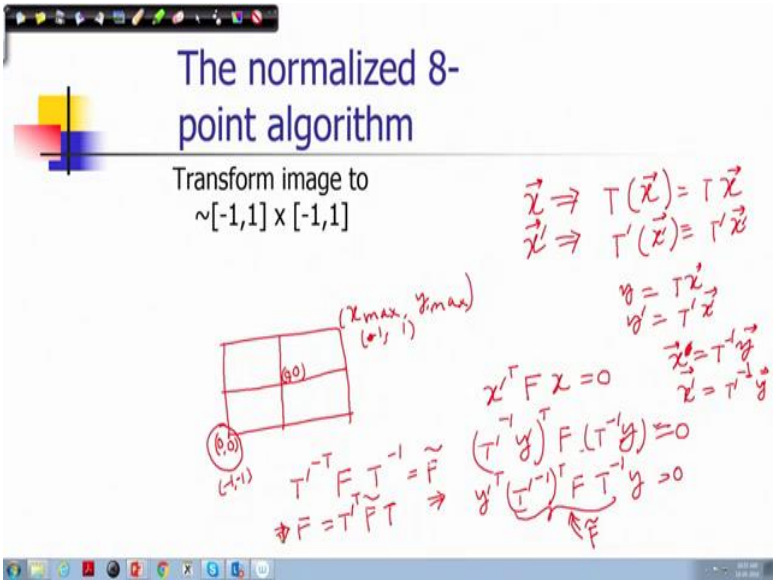
For the right image; let us consider that another transformation we are performing; so we are expressing in this. So, these are linear transformation;

$$(T'x')^T F(Tx) = 0$$

So, if I consider the equation of the fundamental matrix; involving fundamental matrix; so, this is the equation that we are getting; so now you consider apply this transformation; so this is T'.

So, what I will do? I will rather write see transformed point; if the transform point is y. So, I will have y =T x' and y'=T'x' so $x = T^{-1}y$. So, this is a requirement the transformation should be invertible and x prime is T prime inverse y prime; so now this should be written.

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So, now x prime is written as T prime inverse y prime, transpose F and x is T inverse y that is equals 0. And performing this matrix transfer transposition operation; so y prime transpose T prime minus 1 transpose; then F that is T T; T prime inverse transpose, we can write it also T prime minus T in our previous notation; T inverse y equals 0.

Now, you can see this is also a 3 cross 3 matrix and this is the transformed matrix after transformation you get this matrix; you get this one. So, this is the fundamental matrix associated with y transformed points. So, what you can do? You can, you can compute the fundamental matrix between these by using this transformed points; corresponding transformed points where all these coordinates in this ranges will be used. And then from F; F tilde from this whatever you get; then again transform it back to get the fundamental matrix to get the original fundamental matrix; that relationship also we can use. Say this is T prime transpose then F T inverse equals F tilde.

So, now we will multiply at the both end; so both end we will multiply with T. So, it become identity and in this case it would be T prime transpose. So, that would be equal to F. So, this is how you can get; get back the original fundamental matrix. So, with this let me stop here; we will continue this discussion of estimation of fundamental matrix in my next lecture.

Thank you very much.

Keywords: Fundamental matrix, least square solution, normalized algorithm